

MODULE 2

3.2 THE NATURE OF DATA IN ANALYTICS

Data is the main ingredient for any BI, data science, and business analytics initiative. In fact, it can be viewed as the raw material for what these popular decision technologies produce—information, insight, and **knowledge**. Without data none of these technologies could exist and be popularized—although traditionally we have built analytics models using expert knowledge and experience coupled with very little or no data at all; however, those were the old days, and now data is of the essence. Once perceived as a big challenge to collect, store, and manage, data nowadays is widely considered among the most valuable assets of an organization, with the potential to create invaluable insight to better understand customers, competitors, and the business processes.

Data can be small, or it can be very large. It can be structured (nicely organized for computers to process), or it can be unstructured (e.g., text that is created for humans and hence not readily understandable/consumable by computers). It can come in smaller batches continuously or it can pour in all at once as a large batch. These are some of the characteristics that define the inherent nature of today's data, which we often call Big Data. Even though these characteristics of data make it more challenging to process and consume, it also makes it more valuable because it enriches the data beyond its conventional limits, allowing for the discovery of new and novel knowledge. Traditional ways to manually collect data (either via surveys or via human-entered business transactions) mostly left their places to modern-day data collection mechanisms that use Internet and/or sensor/RFID-based computerized networks. These automated data collection systems are not only enabling us to collect more volumes of data but also enhancing the **data quality** and integrity. Figure 3.1 illustrates a typical analytics continuum—data to analytics to actionable information.

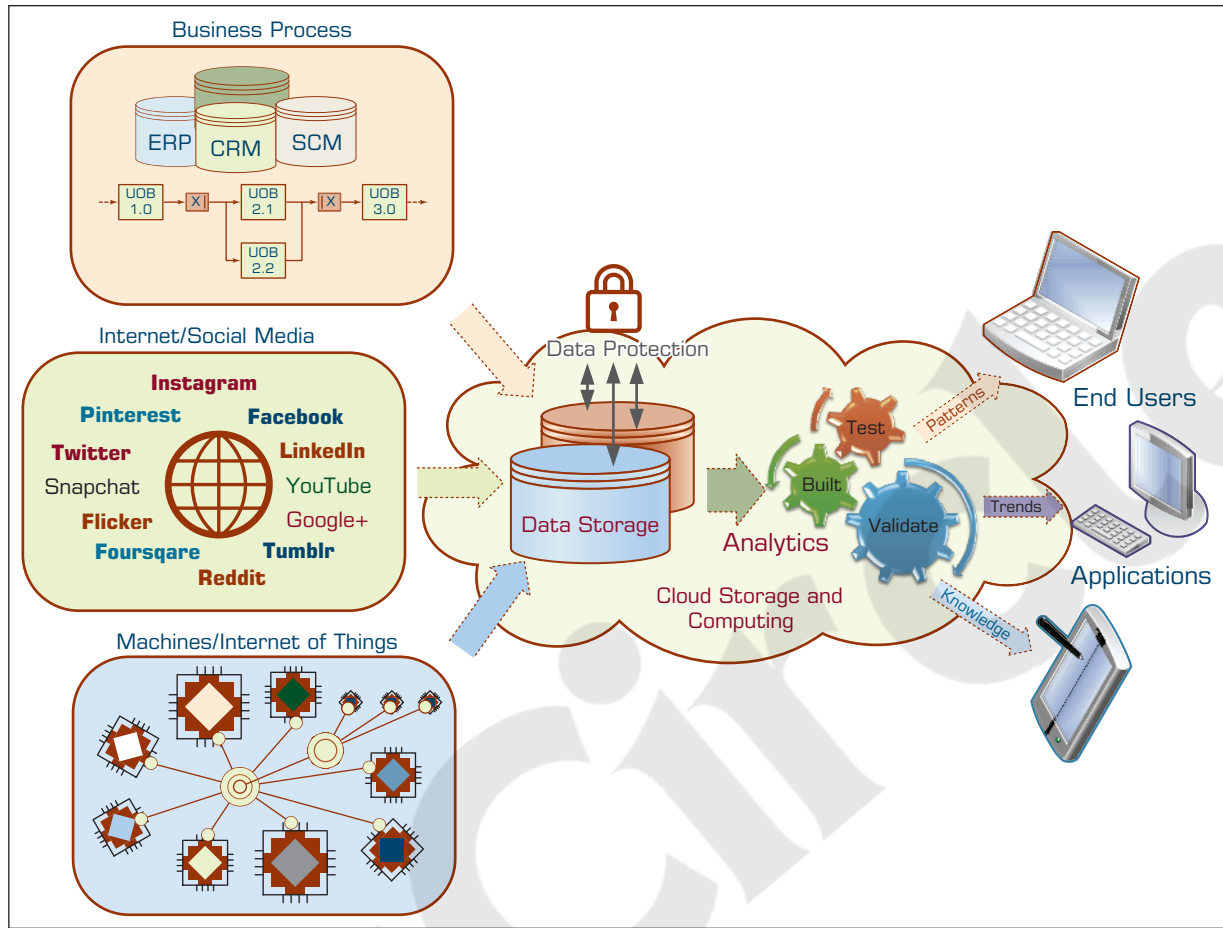


FIGURE 3.1 A Data to Knowledge Continuum.

Although its value proposition is undeniable, to live up its promise, the data must comply with some basic usability and quality metrics. Not all data are useful for all tasks, obviously. That is, data have to match with (have the coverage of the specifics for) the task for which it is intended to be used. Even for a specific task, the relevant data on hand need to comply with the quality and quantity requirements. Essentially, data have to be analytics ready. So what does it mean to make data analytics ready? In addition to its relevancy to the problem at hand and the quality/quantity requirements, it also has to have a certain data structure in place with key fields/variables with properly normalized values. Furthermore, there must be an organization-wide agreed-on definition for common variables and subject matters (sometimes also called high-level data management), such as how you define a customer (what characteristics of customers are used to produce a holistic enough representation to analytics) and where in the business process the customer-related information is captured, validated, stored, and updated.

Sometimes the representation of the data may depend on the type of analytics being employed. Predictive algorithms generally require a flat file with a target variable, so making data **analytics ready** for prediction means that data sets must be transformed into a flat-file format and made ready for ingestion into those predictive algorithms. It is also imperative to match the data to the needs and wants of a specific predictive algorithm and/or a software tool—for instance, neural network algorithms require all input variables to be numerically represented (even the nominal variables need to be converted

into pseudo binary numeric variables) and decision tree algorithms do not require such numerical transformation, easily and natively handling a mix of nominal and numeric variables.

Analytics projects that overlook data-related tasks (some of the most critical steps) often end up with the wrong answer for the right problem, and these unintentionally created, seemingly good, answers may lead to inaccurate and untimely decisions. Following are some of the characteristics (metrics) that define the readiness level of data for an analytics study (Delen, 2020; Kock, McQueen, & Corner, 1997).

- **Data source reliability** refers to the originality and appropriateness of the storage medium where the data is obtained—answering the question of “Do we have the right confidence and belief in this data source?” If at all possible, one should always look for the original source/creator of the data to eliminate/mitigate the possibilities of data misrepresentation and data transformation caused by the mishandling of the data as it moved from the source to destination through one or more steps and stops along the way. Every move of the data creates a chance to unintentionally drop or reformat data items, which limits the integrity and perhaps true accuracy of the data set.
- **Data content accuracy** means that data are correct and are a good match for the analytics problem—answering the question of “Do we have the right data for the job?” The data should represent what was intended or defined by the original source of the data. For example, the customer’s contact information recorded in a record within a database should be the same as what the patient said it was. Data accuracy will be covered in more detail in the following subsection.
- **Data accessibility** means that the data are easily and readily obtainable—answering the question of “Can we easily get to the data when we need to?” Access to data may be tricky, especially if the data is stored in more than one location and storage medium and need to be merged/transformed while accessing and obtaining it. As the traditional relational database management systems leave their place (or coexist with) a new generation of data storage mediums like data lakes and Hadoop infrastructure, the importance/criticality of data accessibility is also increasing.
- **Data security and data privacy** means that the data is secured to only allow those people who have the authority and the need to access it and to prevent anyone else from reaching it. Increasing popularity in educational degrees and certificate programs for Information Assurance is an evidence to the criticality and the increasing urgency of this data quality metric. Any organization that maintains health records for individual patients must have systems in place that not only safeguard the data from unauthorized access (which is mandated by federal laws like Health Insurance Portability and Accountability Act [HIPPA]) but also accurately identifies each patient to allow proper and timely access to records by authorized users (Annas, 2003).
- **Data richness** means that all the required data elements are included in the data set. In essence, richness (or comprehensiveness) means that the available variables portray a rich enough dimensionality of the underlying subject matter for an accurate and worthy analytics study. It also means that the information content is complete (or near complete) to build a predictive and/or prescriptive analytics model.
- **Data consistency** means that the data are accurately collected and combined/merged. Consistent data represent the dimensional information (variables of interest) coming from potentially disparate sources but pertaining to the same subject. If the data integration/merging is not done properly, some of the variables of different subjects may find themselves in the same record—having two different patient records mixed up—for instance, it may happen while merging the demographic and clinical test result data records.

- **Data currency/data timeliness** means that the data should be up-to-date (or as recent/new as it needs to be) for a given analytics model. It also means that the data is recorded at or near the time of the event or observation so that the time-delay-related misrepresentation (incorrectly remembering and encoding) of the data is prevented. Because accurate analytics rely on accurate and timely data, an essential characteristic of analytics-ready data is the timeliness of the creation and access to data elements.
- **Data granularity** requires that the variables and data values be defined at the lowest (or as low as required) level of detail for the intended use of the data. If the data is aggregated, it may not contain the level of detail needed for an analytics algorithm to learn how to discern different records/cases from one another. For example, in a medical setting, numerical values for laboratory results should be recorded to the appropriate decimal place as required for the meaningful interpretation of test results and proper use of those values within an analytics algorithm. Similarly, in the collection of demographic data, data elements should be defined at a granular level to determine the differences in outcomes of care among various subpopulations. One thing to remember is that the data that is aggregated cannot be disaggregated (without access to the original source), but it can easily be aggregated from its granular representation.
- **Data validity** is the term used to describe a match/mismatch between the actual and expected data values of a given variable. As part of data definition, the acceptable values or value ranges for each data element must be defined. For example, a valid data definition related to gender would include three values: male, female, and other (non-binary, transgender, and unknown).
- **Data relevancy** means that the variables in the data set are all relevant to the study being conducted. Relevancy is not a dichotomous measure (whether a variable is relevant or not); rather, it has a spectrum of relevancy from least relevant to most relevant. Based on the analytics algorithms being used, one may choose to include only the most relevant information (i.e., variables) or if the algorithm is capable enough to sort them out, may choose to include all the relevant ones, regardless of their relevancy level. One thing that analytics studies should avoid is to include totally irrelevant data into the model building, as this may contaminate the information for the algorithm, resulting in inaccurate and misleading results.

Although these are perhaps the most prevailing metrics to keep up with, the true data quality and excellent analytics readiness for a specific application domain would require different levels of emphasis paid on these metric dimensions and perhaps add more specific ones to this collection. The following section will dive into the nature of data from a taxonomical perspective to list and define different data types as they relate to different analytics projects.

► SECTION 3.2 REVIEW QUESTIONS

1. How do you describe the importance of data in analytics? Can we think of analytics without data?
2. Considering the new and broad definition of business analytics, what are the main inputs and outputs to the analytics continuum?
3. Where does the data for business analytics come from?
4. In your opinion, what are the top three data-related challenges for better analytics?
5. What are the most common metrics that make for analytics-ready data?

3.3 A SIMPLE TAXONOMY OF DATA

Data (**datum** in singular form) refers to a collection of facts usually obtained as the result of experiments, observations, transactions, or experiences. Data may consist of numbers, letters, words, images, voice recordings, and so on, as measurements of a set of variables (characteristics of the subject or event that we are interested in studying). Data are often viewed as the lowest level of abstraction from which information and then knowledge is derived.

At the highest level of abstraction, one can classify data as structured and unstructured (or semistructured). **Unstructured data**/semistructured data is composed of any combination of textual, imagery, voice, and Web content. Unstructured/semistructured data will be covered in more detail in the text mining and Web mining chapter. **Structured data** is what data mining algorithms use and can be classified as categorical or numeric. The categorical data can be subdivided into nominal or ordinal data, whereas numeric data can be subdivided into intervals or ratios. Figure 3.2 shows a simple **data taxonomy**.

- **Categorical data** represent the labels of multiple classes used to divide a variable into specific groups. Examples of categorical variables include race, sex, age group, and educational level. Although the latter two variables may also be considered in a numerical manner by using exact values for age and highest grade completed, it is often more informative to categorize such variables into a relatively small number of ordered classes. The categorical data may also be called discrete data, implying that it represents a finite number of values with no continuum between them. Even if the values used for the categorical (or discrete) variables are numeric, these numbers are nothing more than symbols and do not imply the possibility of calculating fractional values.
- **Nominal data** contain measurements of simple codes assigned to objects as labels, which are not measurements. For example, the variable *marital status* can be generally categorized as (1) single, (2) married, and (3) divorced. Nominal data can be represented with binomial values having two possible values (e.g., yes/no, true/false, good/bad), or multinomial values having three or more possible values (e.g., brown/green/blue, white/black/Latinx/Asian, single/married/divorced/legal-unions).

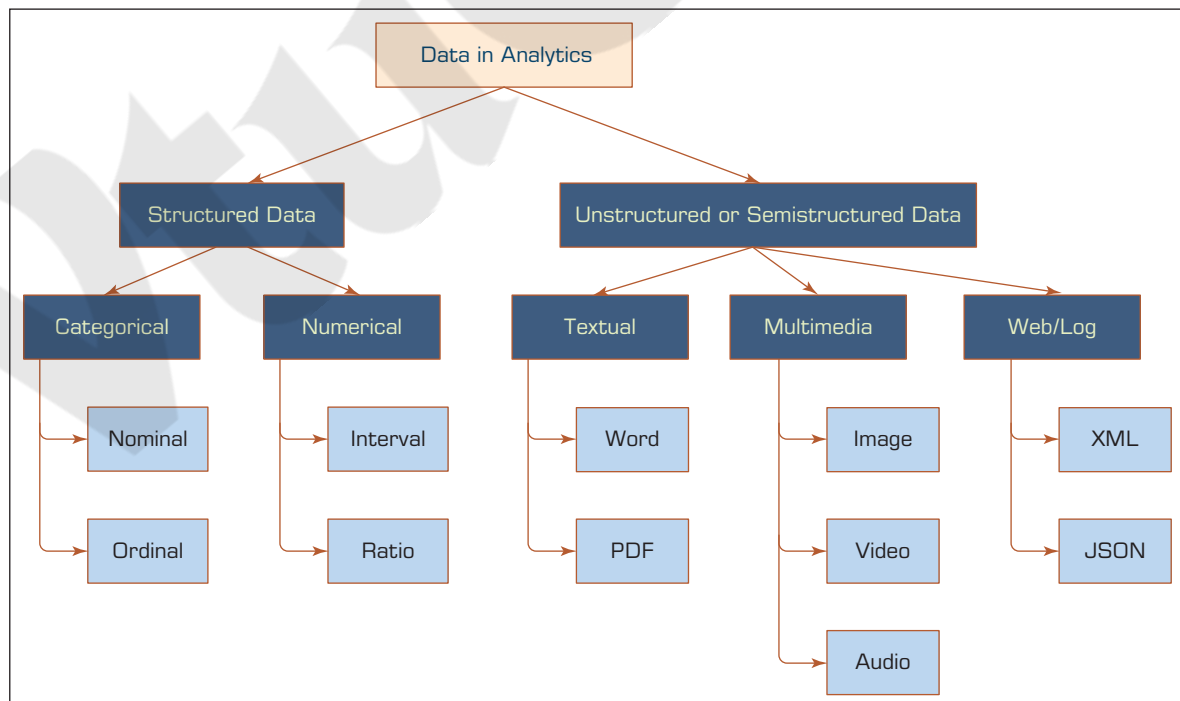


FIGURE 3.2 A Simple Taxonomy of Data.

- **Ordinal data** contain codes assigned to objects or events as labels that also represent the rank order among them. For example, the variable *credit score* can be generally categorized as (1) low, (2) medium, or (3) high. Similar ordered relationships can be seen in variables such as age group (i.e., child, young, middle-aged, older adults) and educational level (i.e., high school, college, graduate school). Some predictive analytic algorithms, such as *ordinal multiple logistic regression*, take into account this additional rank-order information to build a better classification model.
- **Numeric data** represent the numeric values of specific variables. Examples of numerically valued variables include age, number of children, total household income (in U.S. dollars), travel distance (in miles), and temperature (in Fahrenheit degrees). Numeric values representing a variable can be integer (taking only whole numbers) or real (taking also the fractional number). The numeric data may also be called continuous data, implying that the variable contains continuous measures on a specific scale that allows insertion of interim values. Unlike a discrete variable, which represents finite, countable data, a continuous variable represents scalable measurements, and it is possible for the data to contain an infinite number of fractional values.
- **Interval data** are variables that can be measured on interval scales. A common example of interval scale measurement is temperature on the Celsius scale. In this particular scale, the unit of measurement is 1/100 of the difference between the melting temperature and the boiling temperature of water in atmospheric pressure; that is, there is not an absolute zero value.
- **Ratio data** include measurement variables commonly found in the physical sciences and engineering. Mass, length, time, plane angle, energy, and electric charge are examples of physical measures that are ratio scales. The scale type takes its name from the fact that measurement is the estimation of the ratio between a magnitude of a continuous quantity and a unit magnitude of the same kind. Informally, the distinguishing feature of a ratio scale is the possession of a nonarbitrary zero value. For example, the Kelvin temperature scale has a nonarbitrary zero point of absolute zero, which is equal to -273.15 degrees Celsius. This zero point is nonarbitrary because the particles that comprise matter at this temperature have zero kinetic energy.

Other data types, including textual, spatial, imagery, video, and audio/voice, need to be converted into some form of categorical or numeric representation before they can be processed by analytics methods (data mining algorithms; Delen, 2020). Data can also be classified as static or dynamic (i.e., temporal or time series).

Some predictive analytics (i.e., data mining) methods and machine-learning algorithms are very selective about the type of data that they can handle. Providing them with incompatible data types may lead to incorrect models or (more often) halt the model development process. For example, some data mining methods need all the variables (both input as well as output) represented as numerically valued variables (e.g., neural networks, support vector machines, logistic regression). The nominal or ordinal variables are converted into numeric representations using some type of *1-of-N* pseudo variables (e.g., a categorical variable with three unique values can be transformed into three pseudo variables with binary values—1 or 0). Because this process may increase the number of variables, one should be cautious about the effect of such representations, especially for the categorical variables that have large numbers of unique values.

Similarly, some predictive analytics methods, such as ID3 (a classic decision tree algorithm) and rough sets (a relatively new rule induction algorithm), need all the variables represented as categorically valued variables. Early versions of these methods required the user to discretize numeric variables into categorical representations before they could be processed by the algorithm. The good news is that most implementations

of these algorithms in widely available software tools accept a mix of numeric and nominal variables and internally make the necessary conversions before processing the data.

Data comes in many different variable types and representation schemas. Business analytics tools are continuously improving in their ability to help data scientists in the daunting task of data transformation and data representation so that the data requirements of specific predictive models and algorithms can be properly executed.

► SECTION 3.3 REVIEW QUESTIONS

1. What is data? How does data differ from information and knowledge?
2. What are the main categories of data? What types of data can we use for BI and analytics?
3. Can we use the same data representation for all analytics models? Why, or why not?
4. What is a *1-of-N* data representation? Why and where is it used in analytics?

3.4 THE ART AND SCIENCE OF DATA PREPROCESSING

Data in its original form (i.e., the real-world data) is not usually ready to be used in analytics tasks. It is often dirty, misaligned, overly complex, and inaccurate. A tedious and time-demanding process (so-called **data preprocessing**) is necessary to convert the raw real-world data into a well-refined form for analytics algorithms (Kotsiantis, Kanellopoulos, & Pintelas, 2006). Many analytics professionals would testify that the time spent on data preprocessing (which is perhaps the least enjoyable phase in the whole process) is significantly longer than the time spent on the rest of the analytics tasks (the fun of analytics model building and assessment). Figure 3.3 shows the main steps in the data preprocessing endeavor.

In the first phase of data preprocessing, the relevant data is collected from the identified sources, the necessary records and variables are selected (based on an intimate understanding of the data, the unnecessary information is filtered out), and the records coming from multiple data sources are integrated/merged (again, using the intimate understanding of the data, the synonyms and homonyms are able to be handled properly). Such an integration or merging process is nowadays called data blending.

Data blending is a critical part of the most popular job of the 21st century (i.e., data science), including data source blending, data type blending, database blending, time blending, and tool blending (Silipo & Rudnitckaia, 2022). In order to help with wide range of data blending requests, in November 2016, Silipo and her colleagues at KNIME started a blog post series with an intriguing title “Will they Blend?”. In this series, each blog post has been dedicated to a data blending challenge and offered a potential solution. These blog posts were then combined into a book under the same title, *Will They Blend?* Currently, the third edition of the e-book is in existence and can be accessed at (<https://www.knime.com/knimepress/will-they-blend>). The current edition of the e-book contains 32 chapters describing data blending techniques for more than 50 data sources and external tools, from SQL and NoSQL databases to cloud resources, from SharePoint and SAP to Web services and social media, from R and Python scripts to text and images, from MS Word to Web crawling. If you are curious to know whether your data source is covered in the book, just take a look at the topic Index, more than likely, it will be there.

In the second phase of data preprocessing, the data is cleaned (this step is also known as data scrubbing). Data in its original/raw/real-world form is usually dirty (Delen, 2020). In this step, the values in the data set are identified and dealt with. In some cases,

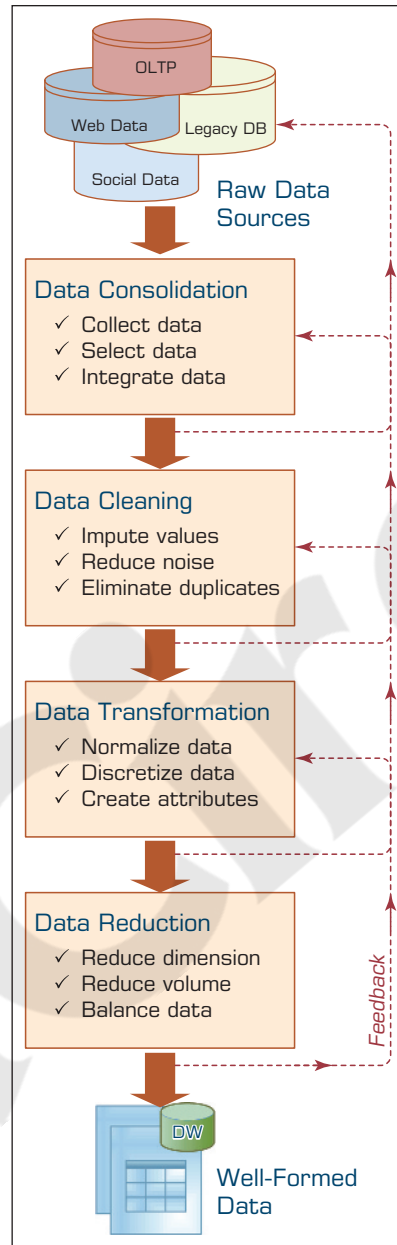


FIGURE 3.3 Data Preprocessing Steps.

missing values are an anomaly in the data set, in which case they need to be imputed (filled with a most probable value) or ignored; in other cases, the missing values are a natural part of the data set (e.g., the *household income* field is often left unanswered by people who are in the top income tier). In this step, the analyst should also identify noisy values in the data (i.e., the outliers) and smooth them out. In addition, inconsistencies (unusual values within a variable) in the data should be handled using domain knowledge and/or expert opinion.

In the third phase of data preprocessing, the data is transformed for better processing. For instance, in many cases the data is normalized between a certain minimum and maximum for all variables to mitigate the potential bias of one variable (having large numeric values, such as for household income) dominating other variables (such as *number of dependents* or *years in service*, which may potentially be more important) having smaller values. Another

transformation that takes place is discretization and/or aggregation. In some cases, the numeric variables are converted to categorical values (e.g., low, medium, high); in other cases, a nominal variable's unique value range is reduced to a smaller set using concept hierarchies (e.g., as opposed to using the individual states with 50 different values, one may choose to use several regions for a variable that shows location) to have a data set that is more amenable to computer processing. Still, in other cases one might choose to create new variables based on the existing ones to magnify the information found in a collection of variables in the data set. For instance, in an organ transplantation data set one might choose to use a single variable showing the blood-type match (1: match, 0: no-match) as opposed to separate multinomial values for the blood type of both the donor and the recipient. Such simplification may increase the information content while reducing the complexity of the relationships in the data.

The final phase of data preprocessing is data reduction. Even though data scientists (i.e., analytics professionals) like to have large data sets, too much data may also be a problem. In the simplest sense, one can visualize the data commonly used in predictive analytics projects as a flat file consisting of two dimensions: variables (the number of columns) and cases/records (the number of rows). In some cases (e.g., image processing and genome projects with complex microarray data), the number of variables can be rather large, and the analyst must reduce the number down to a manageable size. Because the variables are treated as different dimensions that describe the phenomenon from different perspectives, in predictive analytics and data mining this process is commonly called **dimensional reduction** (or **variable selection**). Even though there is not a single best way to accomplish this task, one can use the findings from previously published literature; consult domain experts; run appropriate statistical tests (e.g., principal component analysis or independent component analysis); and, more preferably, use a combination of these techniques to successfully reduce the dimensions in the data into a more manageable and most relevant subset.

With respect to the other dimension (i.e., the number of cases), some data sets may include millions or billions of records. Even though computing power is increasing exponentially, processing such a large number of records may not be practical or feasible. In such cases, one may need to sample a subset of the data for analysis. The underlying assumption of sampling is that the subset of the data will contain all relevant patterns of the complete data set. In a homogeneous data set, such an assumption may hold well, but real-world data is hardly ever homogeneous. The analyst should be extremely careful in selecting a subset of the data that reflects the essence of the complete data set and is not specific to a subgroup or subcategory. The data is usually sorted on some variable, and taking a section of the data from the top or bottom may lead to a biased data set on specific values of the indexed variable; therefore, always try to randomly select the records on the sample set. For skewed data, straightforward random sampling may not be sufficient, and stratified sampling (a proportional representation of different subgroups in the data is represented in the sample data set) may be required. Speaking of skewed data: it is a good practice to balance the highly skewed data by either oversampling the less represented or undersampling the more represented classes. Research has shown that balanced data sets tend to produce better prediction models than unbalanced ones (Thammasiri et al., 2014).

The essence of data preprocessing is summarized in Table 3.1, which maps the main phases (along with their problem descriptions) to a representative list of tasks and algorithms.

It is almost impossible to underestimate the value proposition of data preprocessing. It is one of those time-demanding activities where investment of time and effort pays off without a perceivable limit for diminishing returns. That is, the more resources you invest in it, the more you will gain at the end. Analytics in Action 3.1 illustrates an interesting study where raw, readily available academic data within an educational organization is used to develop predictive models to better understand attrition and improve freshmen student retention in a large higher education institution. As the Analytics in Action clearly states, each and every data preprocessing task described in Table 3.1 was critical to a successful execution of the underlying analytics project, especially the task that related to the balancing of the data set.

TABLE 3.1 A Summary of Data Preprocessing Tasks and Potential Methods

Main Task	Subtasks	Popular Methods
Data consolidation	Access and collect the data	SQL queries, software agents, Web services.
	Select and filter the data	Domain expertise, SQL queries, statistical tests.
	Integrate and unify the data	SQL queries, domain expertise, ontology-driven data mapping.
Data cleaning	Handle missing values in the data	Fill in missing values (imputations) with most appropriate values (mean, median, min/max, mode, etc.); recode the missing values with a constant such as “ML”; remove the record of the missing value; do nothing.
	Identify and reduce noise in the data	Identify the outliers in data with simple statistical techniques (such as averages and standard deviations) or with cluster analysis; once identified, either remove the outliers or smooth them by using binning, regression, or simple averages.
	Find and eliminate erroneous data	Identify the erroneous values in data (other than outliers), such as odd values, inconsistent class labels, odd distributions; once identified, use domain expertise to correct the values or remove the records holding the erroneous values.
Data transformation	Normalize the data	Reduce the range of values in each numerically valued variable to a standard range (e.g., 0 to 1 or −1 to +1) by using a variety of normalization or scaling techniques.
	Discretize or aggregate the data	If needed, convert the numeric variables into discrete representations using range- or frequency-based binning techniques; for categorical variables, reduce the number of values by applying proper concept hierarchies.
	Construct new attributes	Derive new and more informative variables from the existing ones using a wide range of mathematical functions (as simple as addition and multiplication or as complex as a hybrid combination of log transformations).
Data reduction	Reduce number of attributes	Principal component analysis, independent component analysis, chi-square testing, correlation analysis, and decision tree induction.
	Reduce number of records	Random sampling, stratified sampling, expert-knowledge-driven purposeful sampling.
	Balance skewed data	Oversample the less represented or undersample the more represented classes.

ANALYTICS IN ACTION 3.1 Improving Student Retention with Data-Driven Analytics

Student attrition has become one of the most challenging problems for decision makers in academic institutions. Despite all the programs and services that are put in place to help retain students, according to the U.S. Department of Education, Center for Educational Statistics (nces.ed.gov), only about half of those who enter higher education actually earn a bachelor’s degree. Enrolment management and the retention of students has become a top priority for administrators of colleges and universities in the United States and other countries around the world. High dropout of students usually results in overall financial loss, lower graduation rates, and inferior school reputation in the eyes of all stakeholders. The legislators and policy makers who oversee higher education and allocate funds, the parents who pay for their children’s education to prepare them for a better future, and the students who make college choices look for evidence of institutional quality and reputation to guide their decision-making processes.

Proposed Solution

To improve student retention, one should try to understand the nontrivial reasons behind the attrition. To be successful, one should also be able to accurately identify those students that are at risk of dropping out. So far, the vast majority of student attrition research has been devoted

to understanding this complex, yet crucial, social phenomenon. Even though these qualitative, behavioral, and survey-based studies revealed invaluable insight by developing and testing a wide range of theories, they do not provide the much-needed instruments to accurately predict (and potentially improve) student attrition. The project summarized in this case study proposed a quantitative research approach where the historical institutional data from student databases could be used to develop models that are capable of predicting as well as explaining the institution-specific nature of the attrition problem. The proposed analytics approach is shown in Figure 3.4.

Although the concept is relatively new to higher education, for more than a decade now, similar problems in the field of marketing management have been studied using predictive data analytics techniques under the name of “churn analysis,” where the purpose has been to identify among the current customers to answer the question, “Who among our current customers are more likely to stop buying our products or services?” so that some kind of mediation or intervention process can be executed to retain them. Retaining existing customers is crucial because as we all know, and as the related research has shown time and time again, acquiring a new customer costs on an order of magnitude more effort, time, and money than trying to keep the one that you already have.

Data Is of the Essence

The data for this research project came from a single institution (a comprehensive public university located in the Midwest region of the United States) with an average enrollment of 23,000 students, of which roughly 80% are the residents of the same state and roughly 19% of the students are listed as non-white classification. There is no significant difference among genders in the enrollment numbers. The average first-year student retention rate for the institution was about 80%, and the average six-year graduation rate was about 60%.

The study used five years of institutional data, which entailed to 16,000+ students enrolled as first-year students, consolidated from various and diverse university student databases. The data contained variables related to students’ academic, financial, and demographic characteristics. After merging and converting the multidimensional student data into a single flat file (a file with columns representing the variables and rows representing the student records), the resultant file was assessed and preprocessed to identify and remedy anomalies and unusable values. As an example, the study removed all international student records from the data set because they did not contain information about some of the most reputed predictors (e.g., high school GPA, SAT scores). In the data transformation phase, some of the variables were aggregated (e.g., “Major” and “Concentration” variables aggregated to binary variables *MajorDeclared* and *ConcentrationSpecified*) for better interpretation for the predictive modeling. In addition, some of the variables were used to derive new variables (e.g., *Earned/Registered* ratio and *YearsAfterHighSchool*).

$$\text{Earned/Registered} = \text{EarnedHours} / \text{RegisteredHours}$$

$$\text{YearsAfterHighSchool} = \text{FreshmenEnrollmentYear} - \text{HighSchoolGraduationYear}$$

The *Earned/Registered* ratio was created to have a better representation of the students’ resiliency and determination in their first semester of the first year. Intuitively, one would expect greater values for this variable to have a positive impact on retention/persistence. The *YearsAfterHighSchool* was created to measure the impact of the time taken between high school graduation and initial college enrollment. Intuitively, one would expect this variable to be a contributor to the prediction of attrition. These aggregations and derived variables are determined based on a number of experiments conducted for a number of logical hypotheses. The ones that made more common sense and the ones that led to better prediction accuracy were kept in the final variable set. Reflecting the true nature of the subpopulation (i.e., the freshmen students), the dependent variable (i.e., “Second Fall Registered”) contained many more *yes* records (~80%) than *no* records (~20%; see Figure 3.5).

Research shows that having such an imbalanced data has a negative impact on model performance. Therefore, the study experimented with the options of using and comparing the results of the same type of models built with the original imbalanced data (biased for the *yes* records) and the well-balanced data.

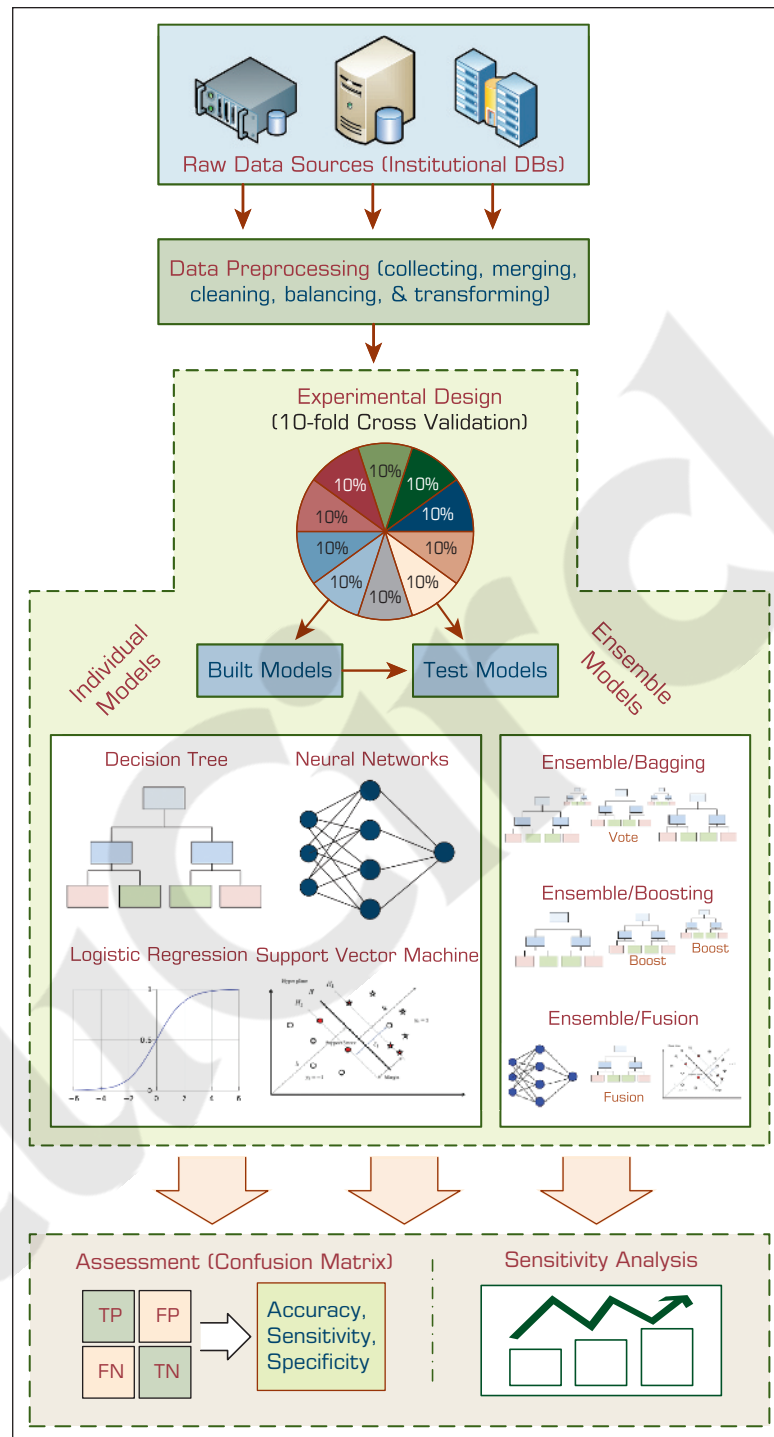


FIGURE 3.4 An Analytics Approach to Predicting Student Attrition.

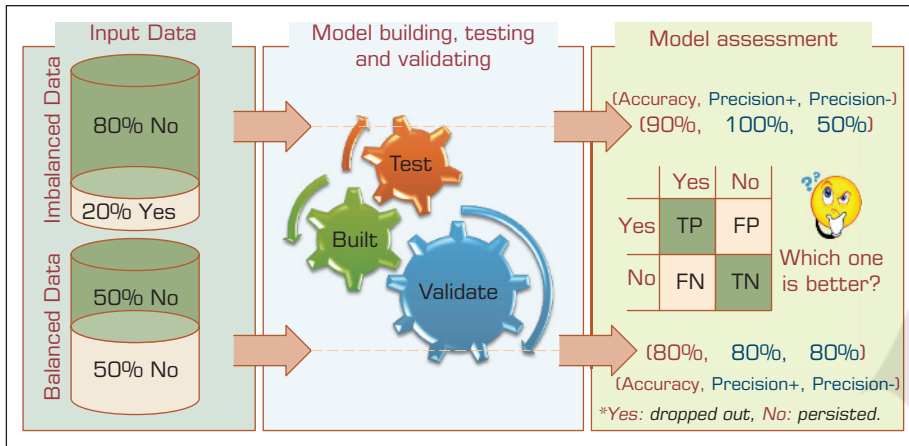


FIGURE 3.5 A Graphical Depiction of the Class Imbalance Problem.

Modeling and Assessment

The study employed four popular classification methods (i.e., artificial neural networks, decision trees, support vector machines, and logistic regression) along with three model ensemble techniques (i.e., bagging, busting, and information fusion). The results obtained from all model types were then compared to each other using regular classification model assessment methods (e.g., overall predictive accuracy, sensitivity, specificity) on the holdout samples.

In machine-learning algorithms (some of which will be covered in Chapter 5), sensitivity analysis is a method for identifying the “cause-and-effect” relationship between the inputs and outputs of a given prediction model. The fundamental idea behind sensitivity analysis is that it measures the importance of predictor variables based on the change in modeling performance that occurs if a predictor variable is not included in the model. This modeling and experimentation practice is also called a leave-one-out assessment. Hence, the measure of sensitivity of a specific predictor variable is the ratio of the error of the trained model without the predictor variable to the error of the model that includes this predictor variable. The more sensitive the network is to a particular variable, the greater the performance decrease would be in the absence of that variable, and therefore the greater the ratio of importance. In addition to the predictive power of the models, the study also conducted sensitivity analyses to determine the relative importance of the input variables.

Results

In the first set of experiments, the study used the original imbalanced data set. Based on the 10-fold cross-validation assessment results, the support vector machines produced the best accuracy with an overall prediction rate of 87.23%, the decision tree came out as the runner-up with an overall prediction rate of 87.16%, followed by artificial neural networks and logistic regression with overall prediction rates of 86.45% and 86.12%, respectively (see Table 3.2).

TABLE 3.2 Prediction Results for the Original/Unbalanced DataSet

	ANN(MLP)		DT(C5)		SVM		LR	
	No	Yes	No	Yes	No	Yes	No	Yes
No	1494	384	1518	304	1478	255	1438	376
Yes	1596	11142	1572	11222	1612	11271	1652	11150
SUM	3090	11526	3090	11526	3090	11526	3090	11526
Per-Class Accuracy	48.35%	96.67%	49.13%	97.36%	47.83%	97.79%	46.54%	96.74%
Overall Accuracy	86.45%		87.16%		87.23%		86.12%	

TABLE 3.3 Prediction Results for the Balanced Data Set

Confusion Matrix	ANN(MLP)		DT(C5)		SVM		LR	
	No	Yes	No	Yes	No	Yes	No	Yes
No	2309	464	2311	417	2313	386	2125	626
Yes	781	2626	779	2673	777	2704	965	2464
SUM	3090	3090	3090	3090	3090	3090	3090	3090
Per-class Accuracy	74.72%	84.98%	74.79%	86.50%	74.85%	87.51%	68.77%	79.74%
Overall Accuracy	79.85%		80.65%		81.18%		74.26%	

A careful examination of these results reveals that the predictions accuracy for the “Yes” class is significantly higher than the prediction accuracy of the “No” class. In fact, all four model types predicted the students who are likely to return for the second year with better than 90% accuracy, but they did poorly on predicting the students who are likely to drop out after the first year with less than 50% accuracy. Because the prediction of the “No” class is the main purpose of this study, less than 50% accuracy for this class was deemed not acceptable. Such a difference in prediction accuracy of the two classes can (and should) be attributed to the imbalanced nature of the training data set (i.e., ~80% “Yes” and ~20% “No” samples).

The next round of experiments used a well-balanced data set where the two classes are represented nearly equally in counts. In realizing this approach, the study took all the samples from the less-represented class (i.e., the “No” class herein) and randomly selected an equal number of samples from the majority class (i.e., the “Yes” class herein) and repeated this process for 10 times to reduce potential bias of random sampling. Each of these sampling processes resulted in a data set of 7,000+ records, of which both class labels (“Yes” and “No”) were equally represented. Again, using a 10-fold cross-validation methodology, the study developed and tested prediction models for all four model types. The results of these experiments are shown in Table 3.3. Based on the holdout sample results, support vector machines once again generated the best overall prediction accuracy with 81.18%, followed by decision trees, artificial neural networks, and logistic regression with an overall prediction accuracy of 80.65%, 79.85%, and 74.26%. As can be seen in the per-class accuracy figures, the prediction models did significantly better on predicting the “No” class with the well-balanced data than they did with the unbalanced data. Overall, the three machine-learning techniques performed significantly better than their statistical counterpart, logistic regression.

Next, another set of experiments were conducted to assess the predictive ability of the three ensemble models. Based on the 10-fold cross-validation methodology, the information fusion-type ensemble model produced the best results with an overall prediction rate of 82.10%, followed by the bagging-type ensembles and boosting-type ensembles with overall prediction rates of 81.80% and 80.21%, respectively (see Table 3.4). Even though the prediction results are slightly better than

TABLE 3.4 Prediction Results for the Three Ensemble Models

	Boosting (Boosted Trees)		Bagging (Random Forest)		Information Fusion (Weighted Average)	
	No	Yes	No	Yes	No	Yes
No	2242	375	2327	362	2335	351
Yes	848	2715	763	2728	755	2739
SUM	3090	3090	3090	3090	3090	3090
Per-Class Accuracy	72.56%	87.86%	75.31%	88.28%	75.57%	88.64%
Overall Accuracy	80.21%		81.80%		82.10%	

the individual models, ensembles are known to produce more robust prediction systems compared to a single-best prediction model (more on this can be found in Chapter 5).

In addition to assessing the prediction accuracy for each model type, a sensitivity analysis was also conducted using the developed prediction models to identify the relative importance of the independent variables (i.e., the predictors). In realizing the overall sensitivity analysis results, each of the four individual model types generated its own sensitivity measures ranking all the independent variables in a prioritized list. As expected, each model type generated slightly different sensitivity rankings of the independent variables. After collecting all four sets of sensitivity numbers, the sensitivity numbers are normalized and aggregated and plotted in a horizontal bar chart (see Figure 3.6).

Conclusions

The study showed that, given sufficient data with the proper variables, data mining methods are capable of predicting freshmen student attrition with approximately 80% accuracy. Results also showed that, regardless of the prediction model employed, the balanced data set (compared

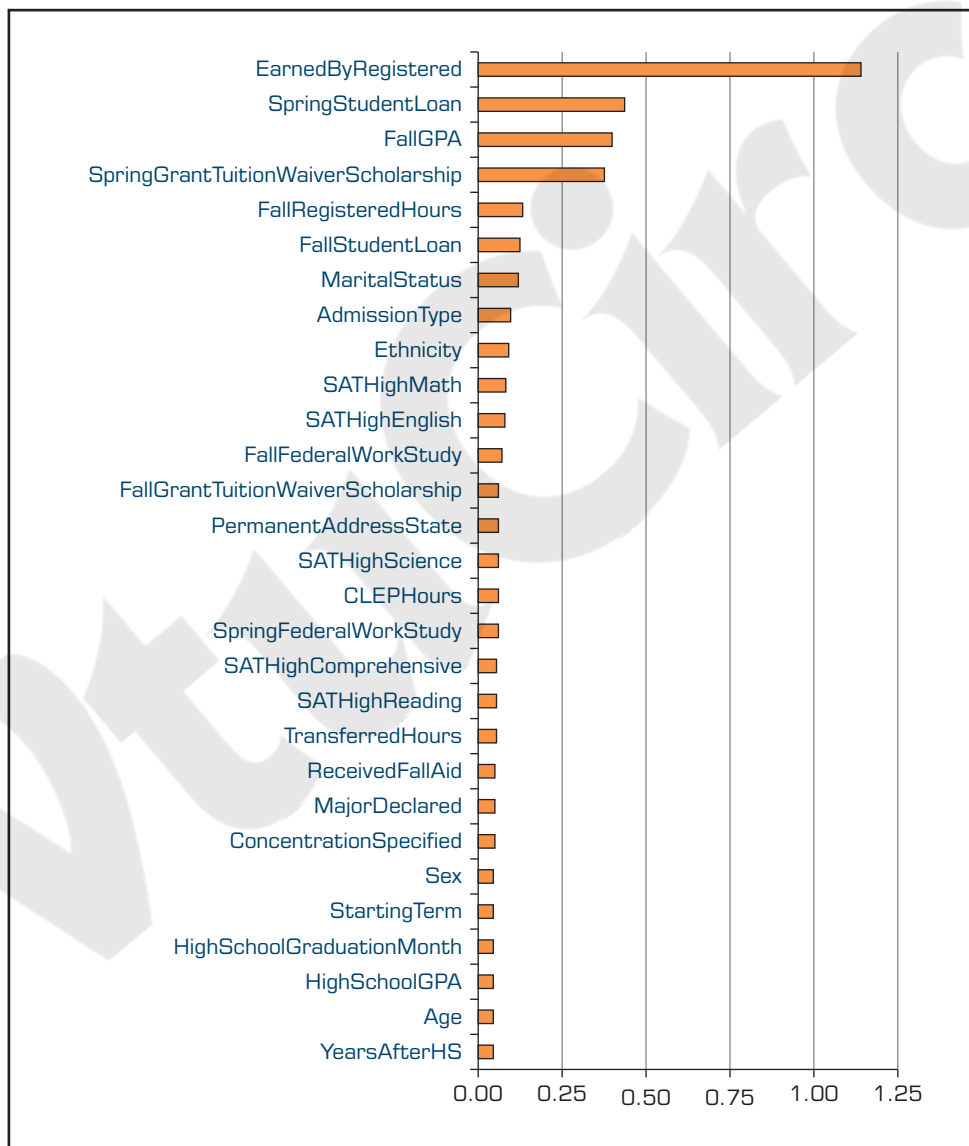


FIGURE 3.6 Sensitivity-Analysis-Based Variable Importance Results.

to unbalanced/original data set) produced better prediction models for identifying the students who are likely to drop out of the college prior to their sophomore year. Among the four individual prediction models used in this study, support vector machines performed the best, followed by decision trees, neural networks, and logistic regression. From the usability standpoint, despite the fact that support vector machines showed better prediction results, one might choose to use decision trees because compared to support vector machines and neural networks, they portray a more transparent model structure. Decision trees explicitly show the reasoning process of different predictions, providing a justification for a specific outcome, whereas support vector machines and artificial neural networks are mathematical models that do not provide such a transparent view of “how they do what they do.”

Sources: Delen, D., Topuz, K., & Eryarsoy, E. (2020). Development of a Bayesian Belief Network-based DSS for predicting and understanding freshmen student attrition. *European Journal of Operational Research*, 281(3), 575–587; Thammasiri, D., Delen, D., Meesad, P., & Kasap N. (2014). A critical assessment of imbalanced class distribution problem: The case of predicting freshmen student attrition. *Expert Systems with Applications*, 41(2), 321–330; Delen, D. (2011). Predicting student attrition with data mining methods. *Journal of College Student Retention*, 13(1), 17–35; Delen, D. (2010). A comparative analysis of machine learning techniques for student retention management. *Decision Support Systems*, 49(4), 498–506.

SECTION 3.4 REVIEW QUESTIONS

1. Why is the original/raw data not readily usable by analytics tasks?
2. What are the main data preprocessing steps?
3. What does it mean to clean/scrub the data? What activities are performed in this phase?
4. Why do we need data transformation? What are the commonly used data transformation tasks?
5. Data reduction can be applied to rows (sampling) and/or columns (variable selection). Which is more challenging?

3.5 DEFINITION OF BIG DATA

Using data to understand customers/clients and business operations to sustain (and foster) growth and profitability is an increasingly challenging task for today's enterprises. As more and more data becomes available in various forms and fashions, timely processing of the data with traditional means becomes impractical. Nowadays, this phenomenon, usually called Big Data is receiving substantial press coverage and drawing increasing interest from both business users and IT professionals. The result is that Big Data is becoming an overhyped and overused marketing buzzword.

Big Data means different things to people with different backgrounds and interests. Traditionally, the term *Big Data* has been used to describe the massive volumes of data analyzed by huge organizations like Google or research science projects at NASA. But for most businesses, it's a relative term: “Big” depends on an organization's size. The point is more about finding new value within and outside conventional data sources. Pushing the boundaries of data analytics uncovers new insights and opportunities, and “big” depends on where you start and how you proceed. Consider the popular description of Big Data: Big Data exceeds the reach of commonly used hardware environments and/or capabilities of software tools to capture, manage, and process it within a tolerable time span for its user population. **Big Data** has become a popular term to describe the exponential growth, availability, and use of information, both structured and unstructured. Much has been written on the Big Data trend and how it can serve as the basis for innovation, differentiation, and growth. Because of the technology challenges in managing the large volume of data coming from multiple sources, sometimes at a rapid speed, additional

new technologies have been developed to overcome the technology challenges. Use of the term *Big Data* is usually associated with such technologies. Because a prime use of storing such data is generating insights through analytics, sometimes the term Big Data is expanded as Big Data analytics. But the term is becoming content free in that it can mean different things to different people. Because our goal is to introduce you to the large data sets and their potential in generating insights, we will use the original term in this chapter.

Where does Big Data come from? A simple answer is “everywhere.” The sources that were ignored because of the technical limitations are now treated as gold mines. Big Data may come from Web logs, radio-frequency identification (RFID), global positioning systems (GPS), sensor networks, social networks, Internet-based text documents, Internet search indexes, detail call records, astronomy, atmospheric science, biology, genomics, nuclear physics, biochemical experiments, medical records, scientific research, military surveillance, photography archives, video archives, and large-scale e-commerce practices.

Big Data is not new. What is new is that the definition and the structure of Big Data constantly change. Companies have been storing and analyzing large volumes of data since the advent of the data warehouses in the early 1990s. Whereas terabytes used to be synonymous with Big Data warehouses, now it’s exabytes, and the rate of growth in data volume continues to escalate as organizations seek to store and analyze greater levels of transaction details, as well as Web- and machine-generated data, to gain a better understanding of customer behavior and business drivers.

Many (academics and industry analysts/leaders alike) think that “Big Data” is a misnomer. What it says and what it means are not exactly the same. That is, Big Data is not just “big.” The sheer volume of the data is only one of many characteristics that are often associated with Big Data, including variety, velocity, veracity, variability, and value proposition, among others.

The “V”s That Define Big Data

Big Data is typically defined by three “V”s: volume, variety, velocity. In addition to these three, we see some of the leading Big Data solution providers adding other “V”s, such as veracity (IBM), variability (SAS), and value proposition.

Volume Volume is obviously the most common trait of Big Data. Many factors contributed to the exponential increase in data volume, such as transaction-based data stored through the years, text data constantly streaming in from social media, increasing amounts of sensor data being collected, automatically generated RFID and GPS data, and so on. In the past, excessive data volume created storage issues, both technical and financial. But with today’s advanced technologies coupled with decreasing storage costs, these issues are no longer significant; instead, other issues have emerged, including how to determine relevance amid the large volumes of data and how to create value from data that is deemed to be relevant.

As mentioned before, *big* is a relative term. It changes over time and is perceived differently by different organizations. With the staggering increase in data volume, even the naming of the next Big Data echelon has been a challenge. The highest mass of data that used to be called petabytes (PB) has left its place to zettabytes (ZB), which is a trillion gigabytes (GB) or a billion terabytes (TB). Technology Insights 3.1 provides an overview of the size and naming of Big Data volumes.

From a short historical perspective, in 2009 the world had about 0.8 ZB of data; in 2010, it exceeded the 1 ZB mark; at the end of 2011, the number was 1.8 ZB. It is expected to be 44 ZB in 2020 (Adshead, 2014). With the growth of sensors and the Internet of Things (IoT), these forecasts could all be wrong. Though these numbers are astonishing in size, so are the challenges and opportunities that come with them.

TECHNOLOGY INSIGHTS 3.1 The Data Size Is Getting Big, Bigger, and Bigger

The metrics of data size is having a hard time keeping up with new names. We all know kilobyte (KB, which is 1,000 bytes), megabyte (MB, which is 1,000,000 bytes), gigabyte (GB, which is 1,000,000,000 bytes), and terabyte (TB, which is 1,000,000,000,000 bytes). Beyond that, the names given to data sizes are relatively new to most of us. The following table shows what comes after terabyte and beyond.

Name	Symbol	Value
Kilobyte	kB	10^3
Megabyte	MB	10^6
Gigabyte	GB	10^9
Terabyte	TB	10^{12}
Petabyte	PB	10^{15}
Exabyte	EB	10^{18}
Zettabyte	ZB	10^{21}
Yottabyte	YB	10^{24}
Brontobyte*	BB	10^{27}
Gegobyte*	GeB	10^{30}

*Not an official SI (International System of Units) name/symbol, yet.

Consider that an exabyte of data is created on the Internet each day, which equates to 250 million DVDs' worth of information. And the idea of even larger amounts of data—a zettabyte—isn't too far off when it comes to the amount of information traversing the Web in any one year. In fact, industry experts are already estimating that we will see 1.3 zettabytes of traffic annually over the Internet by 2016—and it could jump to 2.3 zettabytes by 2020. By 2020, Internet traffic is expected to reach 300 GB per capita per year. When referring to yottabytes, some of the Big Data scientists often wonder about how much data the NSA or FBI have on people altogether. Put in terms of DVDs, a yottabyte would require 250 trillion of them. A brontobyte, which is not an official SI prefix but is apparently recognized by some people in the measurement community, is a 1 followed by 27 zeros. The size of such a magnitude can be used to describe the amount of sensor data that we will get from the Internet in the next decade, if not sooner.

Sources: Backblaze. (2022). What is a yottabyte? at <https://www.backblaze.com/blog/what-is-a-yottabyte/> (accessed May 2022); Byte. (2022). Unit of information, at <https://en.wikipedia.org/wiki/Byte> (accessed April 2022).

Variety Data today come in all types of formats—ranging from traditional databases to hierarchical data stores created by the end users and OLAP systems to text documents, e-mail, XML, meter-collected and sensor-captured data, to video, audio, and stock ticker data. By some estimates, 80 to 85% of all organizations' data are in some sort of unstructured or semistructured format (a format that is not suitable for traditional database schemas). But there is no denying its value, and hence, it must be included in the analyses to support decision making.

Velocity According to Gartner, velocity means both how fast data is being produced and how fast the data must be processed (i.e., captured, stored, and analyzed) to meet the need or demand. RFID tags, automated sensors, GPS devices, and smart meters are driving an increasing need to deal with torrents of data in near real time. Velocity is

perhaps the most overlooked characteristic of Big Data. Reacting quickly enough to deal with velocity is a challenge to most organizations. For time-sensitive environments, the opportunity cost clock of the data starts ticking the moment the data is created. As time passes, the value proposition of the data degrades and eventually becomes worthless. Whether the subject matter is the health of a patient, the well-being of a traffic system, or the health of an investment portfolio, accessing the data and reacting faster to the circumstances will always create more advantageous outcomes.

In the Big Data storm that we are currently witnessing, almost everyone is fixated on at-rest analytics, using optimized software and hardware systems to mine large quantities of variant data sources. Although this is critically important and highly valuable, there is another class of analytics, driven from the velocity of Big Data, called “data stream analytics” or “in-motion analytics,” which is evolving fast. If done correctly, data stream analytics can be as valuable, and in some business environments more valuable, than at-rest analytics. Later in this chapter, we will cover this topic in more detail.

Veracity *Veracity* is a term coined by IBM that is being used as the fourth “V” to describe Big Data. It refers to conformity to facts: accuracy, quality, truthfulness, or trustworthiness of the data. Tools and techniques are often used to handle Big Data’s veracity by transforming the data into quality and trustworthy insights.

Variability In addition to the increasing velocities and varieties of data, data flows can be highly inconsistent with periodic peaks. Is something big trending in the social media? Perhaps there is a high-profile IPO looming. Maybe swimming with manatees in the Bahamas is suddenly the must-do vacation activity. Daily, seasonal, and event-triggered peak data loads can be highly variable and thus challenging to manage—especially with social media involved.

Value Proposition The excitement around Big Data is its value proposition. A preconceived notion about “Big” data is that it contains (or has a greater potential to contain) more patterns and interesting anomalies than “small” data. Thus, by analyzing large and feature-rich data, organizations can gain greater business value that they may not have otherwise. Although users can detect the patterns in small data sets using simple statistical and machine-learning methods or ad hoc query and reporting tools, Big Data means “big” analytics. Big analytics means greater insight and better decisions, something that every organization needs.

Because the exact definition of Big Data (or its successor terms) is still a matter of ongoing discussion in academic and industrial circles, it is likely that more characteristics (perhaps more “V”s) are likely to be added to this list. Regardless of what happens, the importance and value proposition of Big Data is here to stay. Figure 3.7 shows a conceptual architecture where Big Data (at the left side of the figure) is converted to business insight through the use of a combination of advanced analytics and delivered to a variety of different users/roles for faster/better decision making.

► SECTION 3.5 REVIEW QUESTIONS

1. Why is Big Data important? What has changed to put it in the center of the analytics world?
2. How do you define Big Data? Why is it difficult to define?
3. Out of the “V”s that are used to define Big Data, in your opinion, which one is the most important? Why?
4. What do you think the future of Big Data will be like? Will it leave its popularity to something else? If so, what will it be?

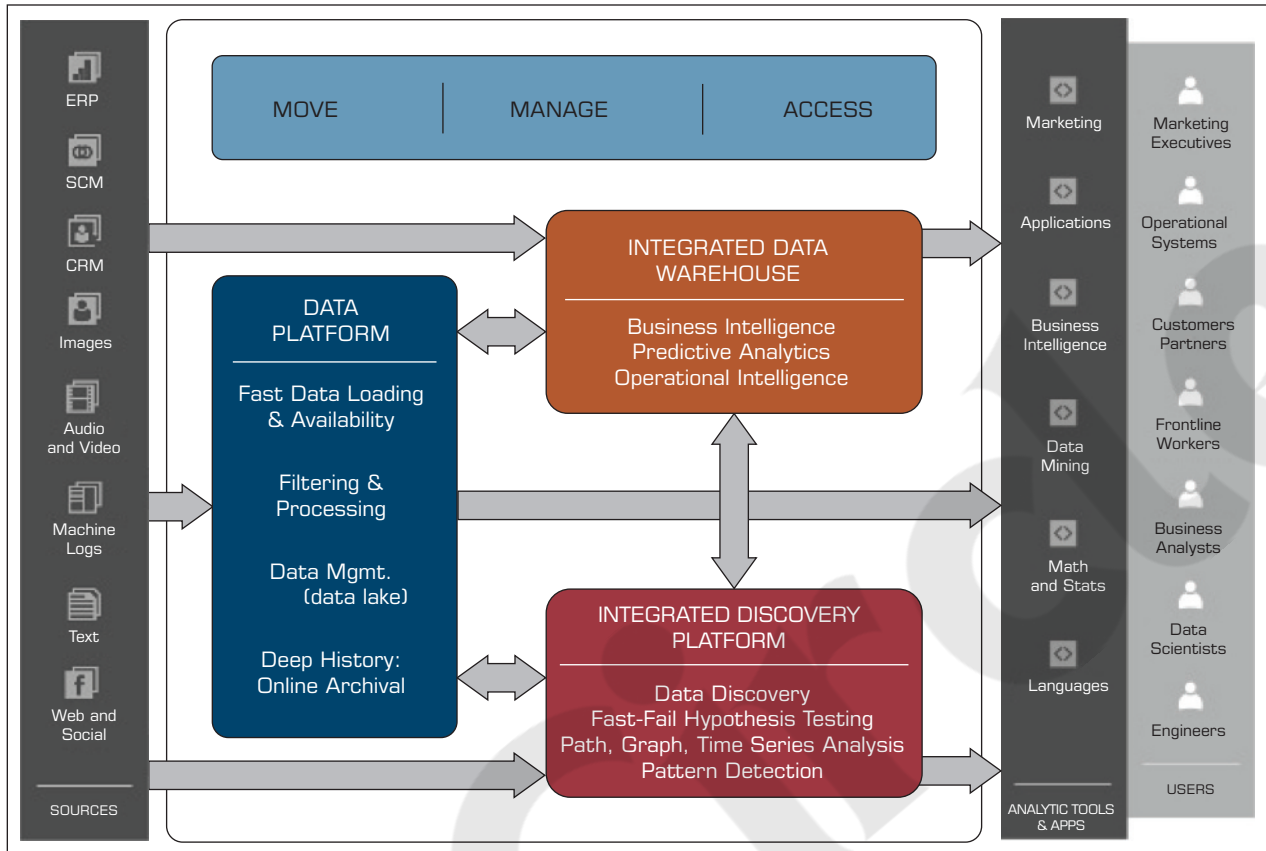


FIGURE 3.7 A High-Level Conceptual Architecture for Big Data Solutions. Source: AsterData—A Teradata Company.

3.6 FUNDAMENTALS OF BIG DATA ANALYTICS

Big Data by itself, regardless of the size, type, or speed, is worthless unless business users do something with it that delivers value to their organizations. That's where "big" analytics comes into the picture. Although organizations have always run reports and dashboards against data warehouses, most have not opened these repositories to in-depth on-demand exploration. This is partly because analysis tools are too complex for the average user but also because the repositories often do not contain all the data needed by the power user. But this is about to change (and has been changing, for some) in a dramatic fashion, thanks to the new Big Data analytics paradigm.

With the value proposition, Big Data also brought about big challenges for organizations. The traditional means for capturing, storing, and analyzing data are not capable of dealing with Big Data effectively and efficiently. Therefore, new breeds of technologies need to be developed (or purchased/hired/outsourced) to take on the Big Data challenge. Before making such an investment, organizations should justify the means. Here are some questions that may help shed light on this situation. If any of the following statements are true, then you need to seriously consider embarking on a Big Data journey.

- You can't process the amount of data that you want to because of the limitations posed by your current platform or environment.
- You want to involve new/contemporary data sources (e.g., social media, RFID, sensory, Web, GPS, textual data) into your analytics platform, but you can't because it

does not comply with the data storage schema-defined rows and columns without sacrificing fidelity or the richness of the new data.

- You need to (or want to) integrate data as quickly as possible to be current on your analysis.
- You want to work with a schema-on-demand (as opposed to predetermined schema used in relational database management systems [RDBMSs]) data storage paradigm because the nature of the new data may not be known, or there may not be enough time to determine it and develop schema for it.
- The data is arriving so fast at your organization's doorstep that your traditional analytics platform cannot handle it.

As is the case with any other large IT investment, the success in Big Data analytics depends on a number of factors. Figure 3.8 shows a graphical depiction of the most critical success factors.

The following are the most critical success factors for **Big Data analytics**:

1. **A clear business need (alignment with the vision and the strategy).** Business investments ought to be made for the good of the business, not for the sake of mere technology advancements. Therefore, the main driver for Big Data analytics should be the needs of the business, at any level—strategic, tactical, and operations.
2. **Strong, committed sponsorship (executive champion).** It is a well-known fact that if you don't have strong, committed executive sponsorship, it is difficult (if not impossible) to succeed. If the scope is a single or a few analytical applications, the sponsorship can be at the departmental level. However, if the target is enterprise-wide organizational transformation, which is often the case for Big Data initiatives, sponsorship needs to be at the highest levels and organization wide.



FIGURE 3.8 Critical Success Factors for Big Data Analytics.

3. **Alignment between the business and IT strategy.** It is essential to make sure that the analytics work is always supporting the business strategy, and not the other way around. Analytics should play the enabling role in successfully executing the business strategy.
4. **A fact-based decision-making culture.** In a fact-based decision-making culture, the numbers rather than intuition, gut feeling, or supposition drive decision making. There is also a culture of experimentation to see what works and what doesn't. To create a fact-based decision-making culture, senior management needs to:
 - Recognize that some people can't or won't adjust
 - Be a vocal supporter
 - Stress that outdated methods must be discontinued
 - Ask to see what analytics went into decisions
 - Link incentives and compensation to desired behaviors
5. **A strong data infrastructure.** Data warehouses have provided the data infrastructure for analytics. This infrastructure is changing and being enhanced in the Big Data era with new technologies. Success requires marrying the old with the new for a holistic infrastructure that works synergistically.

As the size and complexity increase, the need for more efficient analytical systems is also increasing. To keep up with the computational needs of Big Data, a number of new and innovative computational techniques and platforms have been developed. These techniques are collectively called **high-performance computing**, which includes the following:

- **In-memory analytics:** Solves complex problems in near real time with highly accurate insights by allowing analytical computations and Big Data to be processed in-memory and distributed across a dedicated set of nodes.
- **In-database analytics:** Speeds time to insights and enables better data governance by performing data integration and analytic functions inside the database so you won't have to move or convert data repeatedly.
- **Grid computing:** Promotes efficiency, lower cost, and better performance by processing jobs in a shared, centrally managed pool of IT resources.
- **Appliances:** Brings together hardware and software in a physical unit that is not only fast but also scalable on an as-needed basis.

Computational requirements are just a small part of the list of challenges that Big Data impose on today's enterprises. The following is a list of challenges that are found by business executives to have a significant impact on successful implementation of Big Data analytics. When considering Big Data projects and architecture, being mindful of these challenges will make the journey to analytics competency a less stressful one.

Data volume: The ability to capture, store, and process a huge volume of data at an acceptable speed so that the latest information is available to decision makers when they need it.

Data integration: The ability to combine data that is not similar in structure or source and to do so quickly and at a reasonable cost.

Processing capabilities: The ability to process data quickly, as it is captured. The traditional way of collecting and processing data may not work. In many situations, data needs to be analyzed as soon as it is captured to leverage the most value. (This is called *stream analytics*, which will be covered later in this chapter.)

Data governance: The ability to keep up with the security, privacy, ownership, and quality issues of Big Data. As the volume, variety (format and source), and velocity of data change, so should the capabilities of governance practices.

Skills availability: Big Data is being harnessed with new tools and is being looked at in different ways. There is a shortage of people (often called *data scientists*) with skills to do the job.

Solution cost: Because Big Data has opened up a world of possible business improvements, a great deal of experimentation and discovery is taking place to determine the patterns that matter and the insights that turn to value. To ensure a positive return on investment on a Big Data project, therefore, it is crucial to reduce the cost of the solutions used to find that value.

Though the challenges are real, so is the value proposition of Big Data analytics. Anything that you can do as a business analytics leader to help prove the value of new data sources to the business will move your organization beyond experimenting and exploring Big Data into adapting and embracing it as a differentiator. There is nothing wrong with exploration, but ultimately the value comes from putting those insights into action.

Business Problems Addressed by Big Data Analytics

The top business problems addressed by Big Data overall are process efficiency and cost reduction, as well as enhancing customer experience, but different priorities emerge when it is looked at by industry. Process efficiency and cost reduction are perhaps among the top-ranked problems that can be addressed with Big Data analytics for the manufacturing, government, energy and utilities, communications and media, transport, and healthcare sectors. Enhanced customer experience may be at the top of the list of problems addressed by insurance companies and retailers. Risk management usually is at the top of the list for companies in banking and education. Here is a partial list of problems that can be addressed using Big Data analytics:

- Process efficiency and cost reduction
- Brand management
- Revenue maximization, cross-selling, and up-selling
- Enhanced customer experience
- Churn identification, customer recruiting
- Improved customer service
- Identifying new products and market opportunities
- Risk management
- Regulatory compliance
- Enhanced security capabilities

SECTION 3.6 REVIEW QUESTIONS

1. What is Big Data analytics? How does it differ from regular analytics?
2. What are the critical success factors for Big Data analytics?
3. What are the big challenges that one should be mindful of when considering implementation of Big Data analytics?
4. What are the common business problems addressed by Big Data analytics?

3.7 BIG DATA TECHNOLOGIES

There are a number of technologies for processing and analyzing Big Data, but most have some common characteristics (Kelly, 2012). Namely, they take advantage of commodity hardware to enable scale-out and parallel-processing techniques; employ nonrelational data storage capabilities to process unstructured and semistructured data; and apply



Source: Hadoop.

advanced analytics and data visualization technology to Big Data to convey insights to end users. The three Big Data technologies that stand out that most believe will transform the business analytics and data management markets are Hadoop, MapReduce, and NoSQL.

Hadoop

Hadoop is an open-source framework for processing, storing, and analyzing massive amounts of distributed, unstructured data. Originally created by Doug Cutting at Yahoo!, Hadoop was inspired by MapReduce, a user-defined function developed by Google in the early 2000s for indexing the Web. It was designed to handle petabytes and exabytes of data distributed over multiple nodes in parallel. Hadoop clusters run on inexpensive commodity hardware so projects can scale-out without breaking the bank. Hadoop is now a project of the Apache Software Foundation, where hundreds of contributors continuously improve the core technology. Fundamental concept: rather than banging away at one huge block of data with a single machine, Hadoop breaks up Big Data into multiple parts so each part can be processed and analyzed at the same time.

How Does Hadoop Work?

A client accesses structured, unstructured and semistructured data from sources including log files, social media feeds, and internal data stores. Hadoop breaks the data up into smaller “parts,” which are then loaded into a file system made up of multiple nodes running on commodity hardware. The default file store in Hadoop is the **Hadoop Distributed File System**, or **HDFS**. File systems such as HDFS are adept at storing large volumes of unstructured and semistructured data as they do not require data to be organized into relational rows and columns. Each “part” is replicated multiple times and loaded into the file system so that if a node fails, another node has a copy of the data contained on the failed node. A Name Node acts as facilitator, communicating back to the client information such as which nodes are available, where in the cluster certain data resides, and which nodes have failed.

Once the data is loaded into the cluster, it is ready to be analyzed via the MapReduce framework. The client submits a “Map” job—usually a query written in Java—to one of the nodes in the cluster known as the Job Tracker. The Job Tracker refers to the Name Node to determine which data it needs to access to complete the job and where in the cluster that data is located. Once determined, the Job Tracker submits the query to the relevant nodes. Rather than bringing all the data back into a central location for processing, the processing occurs at each node simultaneously, or in parallel. This is an essential characteristic of Hadoop.

When each node has finished processing its given job, it stores the results. The client initiates a “Reduce” job through the Job Tracker in which results of the map phase stored locally on individual nodes are aggregated to determine the “answer” to the original query, and then are loaded onto another node in the cluster. The client accesses these results, which can then be loaded into one of a number of analytic environments for analysis. The MapReduce job has now been completed.

Once the MapReduce phase is complete, the processed data is ready for further analysis by data scientists and others with advanced data analytics skills. **Data scientists** can manipulate and analyze the data using any of a number of tools for any number of uses, including searching for hidden insights and patterns, or use as the foundation for building user-facing analytic applications. The data can also be modeled and transferred from Hadoop clusters into existing relational databases, data warehouses, and other traditional IT systems for further analysis and/or to support transactional processing.

MapReduce

MapReduce is a technique popularized by Google that distributes the processing of very large multistructured data files across a large cluster of machines. High performance is achieved by breaking the processing into small units of work that can be run in parallel across the hundreds, potentially thousands, of nodes in the cluster. To quote the seminal paper on MapReduce:

MapReduce is a programming model and an associated implementation for processing and generating large data sets. Programs written in this functional style are automatically parallelized and executed on a large cluster of commodity machines. This allows programmers without any experience with parallel and distributed systems to easily utilize the resources of a large distributed system. (Maleki et al., 2019)

The key point to note from this quote is that MapReduce is a programming model, not a programming language, that is, it is designed to be used by programmers, rather than business users. The easiest way to describe how MapReduce works is through the use of an example—see the Colored Square Counter in Figure 3.9.

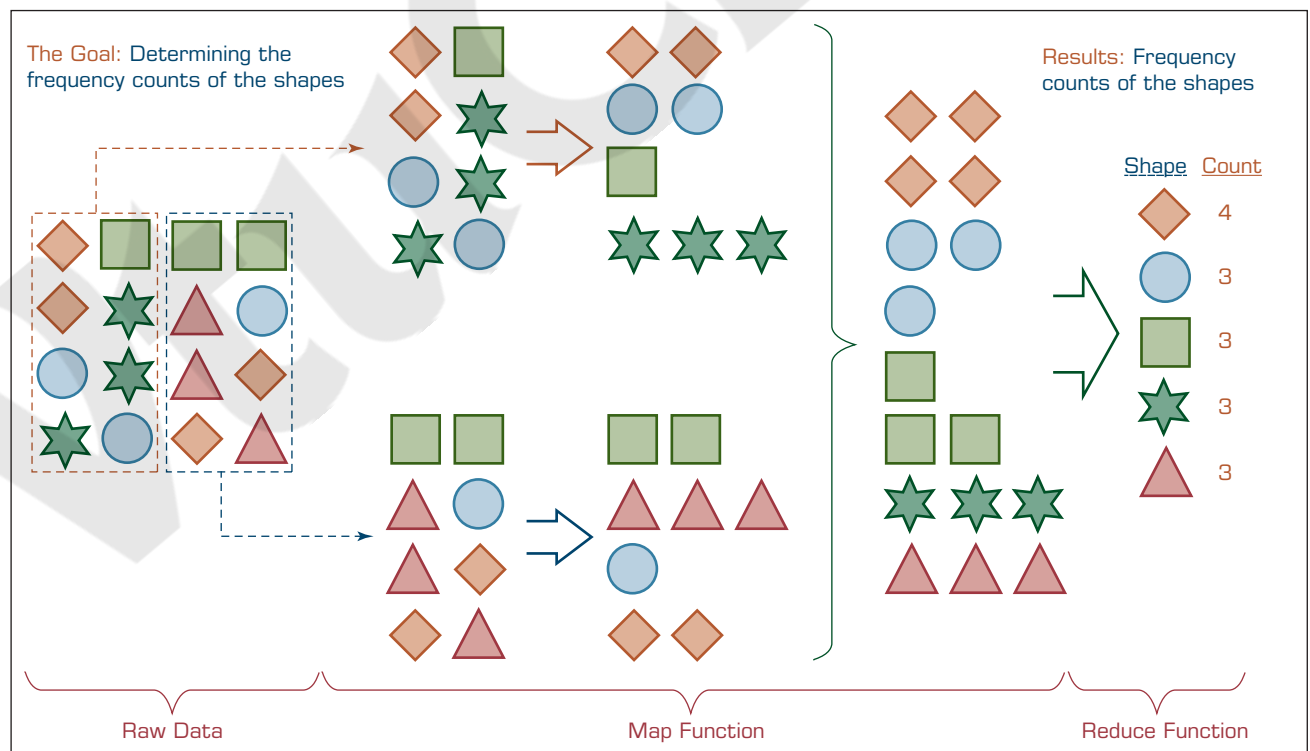


FIGURE 3.9 A Graphical Depiction of the MapReduce Process.

The input to the MapReduce process in Figure 3.9 is a set of colored squares. The objective is to count the number of squares of each color. The programmer in this example is responsible for coding the map and reducing programs; the remainder of the processing is handled by the software system implementing the MapReduce programming model.

The MapReduce system first reads the input file and splits it into multiple pieces. In this example, there are two splits, but in a real-life scenario, the number of splits would typically be much higher. These splits are then processed by multiple map programs running in parallel on the nodes of the cluster. The role of each map program in this case is to group the data in a split by color. The MapReduce system then takes the output from each map program and merges (shuffle/sort) the results for input to the reduce program, which calculates the sum of the number of squares of each color. In this example, only one copy of the reduce program is used, but there may be more in practice. To optimize performance, programmers can provide their own shuffle/sort program and can also deploy a combiner that combines local map output files to reduce the number of output files that have to be remotely accessed across the cluster by the shuffle/sort step.

Why Use MapReduce?

MapReduce aids organizations in processing and analyzing large volumes of multistructured data. Application examples include indexing and search, graph analysis, text analysis, machine learning, data transformation, and so forth. These types of applications are often difficult to implement using the standard SQL employed by relational DBMSs.

The procedural nature of MapReduce makes it easily understood by skilled programmers. It also has the advantage that developers do not have to be concerned with implementing parallel computing—this is handled transparently by the system. Although MapReduce is designed for programmers, nonprogrammers can exploit the value of pre-built MapReduce applications and function libraries. Both commercial and open-source MapReduce libraries are available that provide a wide range of analytic capabilities. Apache Mahout, for example, is an open-source machine-learning library of “algorithms for clustering, classification and batch-based collaborative filtering” that are implemented using MapReduce.

Hadoop Technical Components

In addition to MapReduce, a Hadoop “stack” is made up of a number of components, which include the following:

- **Hadoop Distributed File System (HDFS):** The default storage layer in any given Hadoop cluster.
- **Name Node:** The node in a Hadoop cluster that provides the client information on where in the cluster particular data is stored and if any nodes fail.
- **Secondary Node:** A backup to the Name Node, it periodically replicates and stores data from the Name Node should it fail.
- **Job Tracker:** The node in a Hadoop cluster that initiates and coordinates MapReduce jobs or the processing of the data.
- **Worker Nodes:** The grunts of any Hadoop cluster, worker nodes store data and take direction to process it from the Job Tracker.

In addition to these components, the Hadoop ecosystem is made up of a continuously expanding large number of complementary subprojects. NoSQL data stores like Cassandra and HBase are also used to store the results of MapReduce jobs in Hadoop. In addition to Java, some MapReduce jobs and other Hadoop functions are written in Pig, an open-source language designed specifically for Hadoop. **Hive** is an open-source data

warehouse originally developed by Facebook that allows for analytic modeling within Hadoop. Here are the most commonly referenced subprojects for Hadoop.

Hive Hive is a Hadoop-based data warehousing–like framework originally developed by Facebook. It allows users to write queries in an SQL-like language called HiveQL, which are then converted to MapReduce. This allows SQL programmers with no MapReduce experience to use the warehouse and makes it easier to integrate with business intelligence (BI) and visualization tools such as Microstrategy, Tableau, Revolutions Analytics, and so forth.

Pig **Pig** is a Hadoop-based query language developed by Yahoo! It is relatively easy to learn and is adept at very deep, very long data pipelines (a limitation of SQL).

HBase HBase is a nonrelational database that allows for low-latency, quick lookups in Hadoop. It adds transactional capabilities to Hadoop, allowing users to conduct updates, inserts, and deletes. eBay and Facebook use HBase heavily.

Flume Flume is a framework for populating Hadoop with data. Agents are populated throughout one's IT infrastructure—inside Web servers, application servers, and mobile devices, for example—to collect data and integrate it into Hadoop.

Oozie Oozie is a workflow processing system that lets users define a series of jobs written in multiple languages—such as MapReduce, Pig, and Hive—and then intelligently link them to one another. Oozie allows users to specify, for example, that a particular query is only to be initiated after specified previous jobs on which it relies for data are completed.

Ambari Ambari is a Web-based set of tools for deploying, administering, and monitoring Apache Hadoop clusters. Its development is being led by engineers from Hortonworks, which includes Ambari in its Hortonworks Data Platform.

Avro Avro is a data serialization system that allows for encoding the schema of Hadoop files. It is adept at parsing data and performing removed procedure calls.

Mahout Mahout is a data mining library. It takes the most popular data mining algorithms for performing clustering, regression testing, and statistical modeling and implements them using the MapReduce model.

Sqoop Sqoop is a connectivity tool for moving data from non-Hadoop data stores—such as relational databases and data warehouses—into Hadoop. It allows users to specify the target location inside of Hadoop and instructs Sqoop to move data from Oracle, Teradata, or other relational databases to the target.

HCatalog HCatalog is a centralized metadata management and sharing service for Apache Hadoop. It allows for a unified view of all data in Hadoop clusters and allows diverse tools, including Pig and Hive, to process any data elements without needing to know physically where in the cluster the data is stored.

Hadoop: The Pros and Cons

The main benefit of Hadoop is that it allows enterprises to process and analyze large volumes of unstructured and semistructured data, heretofore inaccessible to them, in a cost- and time-effective manner. Because Hadoop clusters can scale to petabytes and even

exabytes of data, enterprises no longer must rely on sample data sets but can process and analyze *all* relevant data. Data scientists can apply an iterative approach to analysis, continually refining and testing queries to uncover previously unknown insights. It is also inexpensive to get started with Hadoop. Developers can download the Apache Hadoop distribution for free and begin experimenting with Hadoop in less than a day.

The downside to Hadoop and its myriad components is that they are immature and still developing. As with any young, raw technology, implementing and managing Hadoop clusters and performing advanced analytics on large volumes of unstructured data require significant expertise, skill, and training. Unfortunately, there is currently a dearth of Hadoop developers and data scientists available, making it impractical for many enterprises to maintain and take advantage of complex Hadoop clusters. Further, as Hadoop's myriad components are improved on by the community and new components are created, there is, as with any immature open-source technology/approach, a risk of forking. Finally, Hadoop is a batch-oriented framework, meaning it does not support real-time data processing and analysis. In order to increase the efficiency and real-time data processing, another open-source big data technology under the name of Spark is developed by the Apache Foundation, which is explained in the next subsection.

The good news is that some of the brightest minds in IT are contributing to the open-source big data technology development projects, and a new generation of developers and data scientists is coming of age. As a result, the technology is advancing rapidly, becoming both more powerful and easier to implement and manage. An ecosystem of vendors, both recently founded start-ups like Cloudera and Hortonworks and well-known IT stalwarts like IBM, Microsoft, Teradata, and Oracle are working to offer commercial, enterprise-ready big data tools and services to make deploying and managing the underlying technology a practical reality for the traditional enterprise. Several other leading-edge start-ups are working to perfect NoSQL (Not Only SQL) data stores capable of delivering near-real-time insights in conjunction with Hadoop and Spark. Technology Insights 3.2 provides a few facts to clarify some misconceptions about Hadoop.

TECHNOLOGY INSIGHTS 3.2 A Few Demystifying Facts about Hadoop

Although Hadoop and related technologies have been around for more than five years now, most people still have several misconceptions about Hadoop and related technologies such as MapReduce and Hive. The following list of 10 facts intends to clarify what Hadoop is and does relative to BI, as well as in which business and technology situations Hadoop-based BI, data warehousing, and analytics can be useful (Russom, 2013).

Fact #1. Hadoop consists of multiple products. We talk about Hadoop as if it's one monolithic software, whereas it's actually a family of open-source products and technologies overseen by the Apache Software Foundation (ASF). (Some Hadoop products are also available via vendor distributions; more on that later.)

The Apache Hadoop library includes (in BI priority order) HDFS, MapReduce, Hive, Hbase, Pig, Zookeeper, Flume, Sqoop, Oozie, Hue, and so on. You can combine these in various ways, but HDFS and MapReduce (perhaps with Hbase and Hive) constitute a useful technology stack for applications in BI, data warehousing, and analytics.

Fact #2. Hadoop is open source but available from vendors, too. Apache Hadoop's open-source software library is available from ASF at apache.org. For users desiring a more enterprise-ready package, a few vendors now offer Hadoop distributions that include additional administrative tools and technical support.

Fact #3. Hadoop is an ecosystem, not a single product. In addition to products from Apache, the extended Hadoop ecosystem includes a growing list of vendor products that integrate

with or expand Hadoop technologies. One minute on your favorite search engine will reveal these.

Fact #4. HDFS is a file system, not a database management system (DBMS). Hadoop is primarily a distributed file system and lacks capabilities we would associate with a DBMS, such as indexing, random access to data, and support for SQL. That's okay, because HDFS does things DBMSs cannot do.

Fact #5. Hive resembles SQL but is not standard SQL. Many of us are handcuffed to SQL because we know it well and our tools demand it. People who know SQL can quickly learn to hand-code Hive, but that doesn't solve compatibility issues with SQL-based tools. TDWI feels that over time, Hadoop products will support standard SQL, so this issue will soon be moot.

Fact #6. Hadoop and MapReduce are related but don't require each other. Developers at Google developed MapReduce before HDFS existed, and some variations of MapReduce work with a variety of storage technologies, including HDFS, other file systems, and some DBMSs.

Fact #7. MapReduce provides control for analytics, not analytics per se. MapReduce is a general-purpose execution engine that handles the complexities of network communication, parallel programming, and fault tolerance for any kind of application that you can hand code—not just analytics.

Fact #8. Hadoop is about data diversity, not just data volume. Theoretically, HDFS can manage the storage and access of any data type as long as you can put the data in a file and copy that file into HDFS. As outrageously simplistic as that sounds, it's largely true, and it's exactly what brings many users to Apache HDFS.

Fact #9. Hadoop complements a DW; it's rarely a replacement. Most organizations have designed their DW for structured, relational data, which makes it difficult to wring BI value from unstructured and semistructured data. Hadoop promises to complement DWs by handling the multistructured data types most DWs can't.

Fact #10. Hadoop enables many types of analytics, not just Web analytics. Hadoop gets a lot of press about how Internet companies use it for analyzing Web logs and other Web data, but other use cases exist. For example, consider the Big Data coming from sensory devices, such as robotics in manufacturing, RFID in retail, or grid monitoring in utilities. Older analytic applications that need large data samples—such as customer-base segmentation, fraud detection, and risk analysis—can benefit from the additional Big Data managed by Hadoop. Likewise, Hadoop's additional data can expand 360-degree views to create a more complete and granular view.

Spark versus Hadoop

To leverage the value proposition of the Big Data phenomenon, over the last couple of decades, several novel data processing frameworks are proposed. Among the ones proposed to date, the two most commonly referenced and widely used ones have been Hadoop and Spark. Both of these open-source frameworks are developed by the Apache Software Foundation (Hadoop is released in 2004 and Spark is released in 2009) to manage and process modern-day Big Data sets.

Hadoop (or more appropriately Apache Hadoop), as described in more detail above, is a well-known, one of the first successfully developed and deployed framework to cope with Big Data. It can handle not only very large sizes of data (i.e., volume), but also a wide range of data types (i.e., variety) created at a unprecedented speed (i.e., velocity). Key benefits of Hadoop include (1) handling the Big Data with commodity hardware, (2) preventing loss of data and information due to hardware failures through replications, (3) scaling from a small cluster to a very large analytics system, and (4) enabling the discovery of knowledge from Big Data cost effectively and efficiently.

Spark (or more appropriately Apache Spark) is also an open-source Big Data processing system developed to address the challenges of Big Data, and to create value

from the new portrait of the digital world. Similar to Hadoop, Spark also decomposes large data management and analysis tasks into smaller pieces to be handled by large volumes of different nodes built on regular storage and computing hardware devices. The key difference is that Spark uses in-memory processing (i.e., random access memory, or RAM) to cache and analyze the data (instead of a file system, which has been the case in Hadoop) to create result significantly faster than Hadoop. Many believe that the key motivation behind developing Spark several years after Hadoop was the dire need for computational efficiency. Some of the most commonly cited benefits of Spark include (1) order of magnitude faster processing of big data, (2) a unified engine that supports highly efficient SQL queries, streaming data, machine learning and graph processing, and (3) a revamped APIs designed for ease of use, especially for processing of unstructured and semistructured data.

These two frameworks differ in their ecosystems. While Hadoop ecosystem consists of four primary modules (i.e., Hadoop Distributed File System (HDFS), Yet Another Resource Negotiator (YARN), Hadoop MapReduce, Hadoop Common (Hadoop Core)), Spark ecosystem consists of five modules (i.e., Spark Core, Spark SQL, Spark Streaming and Structured Streaming, Machine Learning Library (MLlib), GraphX). Some of the key comparison points between Hadoop and Spark are as follows:

- **Performance:** Spark is faster because it uses random access memory (RAM) instead of reading and writing intermediate data to disks. In contrast, Hadoop stores data on multiple sources and processes it in batches via MapReduce.
- **Cost:** Hadoop runs at a lower cost since it relies on any disk storage type for data processing. Spark runs at a higher cost because it relies on in-memory computations for real-time data processing, which requires it to use high quantities of RAM to spin up nodes.
- **Parallel processing:** Though both platforms process data in parallel in a distributed environment, Hadoop is ideal for batch processing and linear data processing whereas Spark is ideal for real-time processing and processing live unstructured data streams.
- **Scalability:** When data volume rapidly grows, Hadoop quickly scales to accommodate the demand via Hadoop Distributed File System (HDFS). In turn, Spark relies on the fault tolerant HDFS for large volumes of data.
- **Security:** Spark enhances security with authentication via shared secret or event logging, whereas Hadoop uses multiple authentication and access control methods. Though, overall, Hadoop is more secure, Spark can integrate with Hadoop to reach a higher security level.
- **Analytics:** Spark is the superior platform in this category because it includes MLlib, which performs iterative in-memory ML computations. It also includes tools that perform regression, classification, persistence, pipeline construction, evaluation, etc.

There are misconceptions when it comes to comparing Hadoop and Spark. Here are a few most common ones:

- Spark replaced Hadoop. Although there is a partial truth to it, it is not entirely correct. Both Hadoop and Spark have their respective use cases (as mentioned below).
- Hadoop is a database and Spark is an analytics engine. This is not true as both Hadoop and Spark can be used as both data management and data analysis tools.
- Hadoop is cheaper. Although both Hadoop and Spark are free open-source frameworks, professional installation and use of these tools are anything but free, requiring significant investment on consultancy, data scientists, and appropriate hardware infrastructure.

- Spark is always order of magnitude (up to 100 times) faster than Hadoop. That is not true. Although for small data processing tasks Spark can perform 100 times faster than Hadoop, its efficiency diminishes significantly for very large data processing tasks.

So, how do we decide when to use Hadoop and when to use Spark:

Use Hadoop for:

- Processing big data sets in environments where data size exceeds available memory
- Batch processing with tasks that exploit disk read and write operations
- Building data analysis infrastructure with a limited budget
- Completing jobs that are not time-sensitive
- Historical and archive data analysis

Use Spark for:

- Dealing with chains of parallel operations by using iterative algorithms
- Achieving quick results with in-memory computations
- Analyzing stream data analysis in real time
- Graph-parallel processing to model data
- All ML applications

NoSQL

A related new style of database called **NoSQL** (Not Only SQL) has emerged to, like Hadoop, process large volumes of multistructured data. However, whereas Hadoop is adept at supporting large-scale, batch-style historical analysis, NoSQL databases are aimed, for the most part (though there are some important exceptions), at serving up discrete data stored among large volumes of multistructured data to end-user and automated Big Data applications. This capability is sorely lacking from relational database technology, which simply can't maintain needed application performance levels at a Big Data scale.

In some cases, NoSQL and Hadoop work in conjunction. The aforementioned HBase, for example, is a popular NoSQL database modeled after Google BigTable that is often deployed on top of HDFS, the Hadoop Distributed File System, to provide low-latency, quick lookups in Hadoop. The downside of most NoSQL databases today is that they trade ACID (atomicity, consistency, isolation, durability) compliance for performance and scalability. Many also lack mature management and monitoring tools. Both of these shortcomings are in the process of being overcome by the open-source NoSQL communities and a handful of vendors that are attempting to commercialize the various NoSQL databases. NoSQL databases currently available include HBase, Cassandra, MongoDB, Accumulo, Riak, CouchDB, and DynamoDB, among others.

Data for Good

The recent years have witnessed a significant proliferation in volunteered and crowd-sourced community initiatives. Although a significant amount of these initiatives has focused on algorithms (mathematical, statistical, and machine learning), programming languages (e.g., Python and R), and software tools (e.g., KNIME and Orange), some of them have been focusing on solving socially and environmentally meaningful problems and to discover new patterns and knowledge. The latter types of initiatives are often called “data for good” projects. The fundamental idea behind these projects is to do good with data by using mostly publicly available data sources and business analytics and data science tools and techniques to create new understanding, awareness, and solutions to problems that affect the environment, large populations, and often less fortunate groups or minorities. Table 3.5 lists and briefly defines some of the most prolific of these projects.

TABLE 3.5 Data for Good Project Examples

Name of Initiative	Sponsor	Description	URL
TIBCO4Good	TIBCO	An initiative is fueled by a passion for solving worldwide challenges through data. The goal is to make a bigger impact for humanitarian efforts around the world by mobilizing the ecosystem through strategic programs.	https://www.tibco.com/tibco4good .
Data for Good	SAS	SAS is proud to be part of the Data for Good movement, which encourages using data in meaningful ways to solve humanitarian issues around poverty, health, human rights, education, and the environment.	https://www.sas.com/en_us/data-for-good.html
Science for Social Good	IBM	Applying artificial intelligence, cloud, and deep science to scale social impact where applied science can help solve the world's toughest problems and inspire business innovations.	https://research.ibm.com/science-for-social-good/
Doing good with Data	Teradata	Teradata is dedicated to helping non-profit organizations around the globe use data to tackle the problems they face every day.	https://www.teradata.com/About-Us/Doing-Good-With-Data
DataKind	Tableau	An organization that aims to harness the power of data science in the service of humanity, and brings solutions to big social problems. DataKind is considered a leader in the Data for Good movement.	https://www.datakind.org/
Data for Good	A Canadian Foundation	A collective of do gooders, who want to use their powers for good, and not evil, to help make our communities better through data. This is a national not for profit organization, with chapters across the country (i.e., Canada), that help other not for profit, and non-governmental, organizations harness the power of their data to make more informed and better decisions in their quest to make their communities flourish.	https://dataforgood.ca/
Data for Good	Facebook	The goal is to empower partners with privacy-preserving data that strengthens communities and advances social issues. Data for Good is helping organizations respond to the COVID-19 pandemic.	https://dataforgood.facebook.com/
Data & Analytics for Good	Open Journal	Data & Analytics for Good is an open access, peer-reviewed data journal with an international focus that is dedicated to publishing novel research that furthers one or more of the United Nations' sustainable development goals.	https://data-for-good.pubpub.org/
Data for Good	Appsilon	The team behind this initiative has gathered significant expertise in interactive data visualization, machine learning, and developing artificial intelligence solutions and put these technologies to work in resolving the world's most pressing challenges.	https://appsilon.com/data-for-good/
Democratizing Data	Data.org	Data.org is a platform for partnerships to build the field of data science for social impact. We work with organizations from all over the world to increase the use of data science in order to improve the lives of millions of people.	https://data.org/

There are many local Data for Good initiatives. If you are interested in advancing your knowledge and experience in business analytics and data science, it would be a good idea to get involved in some of these community-based volunteered initiatives. That way, while learning analytics, you can also make an impact on environment preservation and improving the wellness of humanity.

SECTION 3.7 REVIEW QUESTIONS

1. What are the common characteristics of emerging Big Data technologies?
2. What is Hadoop? How does it work?
3. What are the main Hadoop components? What functions do they perform?
4. What is MapReduce? What does it do? How does it do it?
5. What is Spark and how does it compare to Hadoop? What are the similarities and differences?
6. What is NoSQL? How does it fit into the Big Data analytics picture?
7. What is Data for Good? Select, study, and report on two of the Data for Good initiatives. In your report, make sure to comment on the origins, motivations, and outcomes of the initiatives.

3.8 BIG DATA AND STREAM ANALYTICS

Along with volume and variety, as we have seen earlier in this chapter, one of the key characteristics that defines Big Data is velocity, which refers to the speed at which the data is created and streamed into the analytics environment. Organizations are looking for new means to process streaming data as it comes in to react quickly and accurately to problems and opportunities to please their customers and to gain a competitive advantage. In situations where data streams in rapidly and continuously, traditional analytics approaches that work with previously accumulated data (i.e., data at rest) often either arrive at the wrong decisions because of using too much out-of-context data, or they arrive at the correct decisions but too late to be of any use to the organization. Therefore, it is critical for a number of business situations to analyze the data soon after it is created and/or as soon as it is streamed into the analytics system.

The presumption that the vast majority of modern-day businesses are currently living by is that it is important and critical to record every piece of data because it might contain valuable information now or sometime in the near future. However, as long as the number of data sources increases, the “store-everything” approach becomes harder and harder and, in some cases, not even feasible. In fact, despite technological advances, current total storage capacity lags far behind the digital information being generated in the world. Moreover, in the constantly changing business environment, real-time detection of meaningful changes in data as well as of complex pattern variations within a given short time window is essential to come up with the actions that better fit with the new environment. These facts become the main triggers for a paradigm that we call *stream analytics*. The stream analytics paradigm was born as an answer to these challenges, namely, the unbounded flows of data that cannot be permanently stored to be subsequently analyzed, in a timely and efficient manner, and complex pattern variations that need to be detected and acted on as soon as they happen.

Stream analytics (also called *data-in-motion analytics* and *real-time data analytics*, among others) is a term commonly used for the analytic process of extracting actionable information from continuously flowing/streaming data. A stream is defined as a continuous sequence of data elements (Zikopoulos et al., 2013). The data elements in a stream are often called *tuples*. In a relational database sense, a tuple is similar to a row of data (a record, an object, an instance). However, in the context of semistructured or unstructured data, a tuple is an abstraction that represents a package of data, which can be characterized as a set of attributes for a given object. If a tuple by itself is not sufficiently informative for analysis or a correlation—or other collective relationships among tuples are needed—then a window of data that includes a set of tuples is used. A window of data is a finite number/sequence of tuples, where the windows are continuously updated

as new data become available. The size of the window is determined based on the system being analyzed. Stream analytics is becoming increasingly more popular because of two things. First, time-to-action has become an ever-decreasing value, and second, we have the technological means to capture and process the data while it is created.

Some of the most impactful applications of stream analytics were developed in the energy industry, specifically for smart grid (electric power supply chain) systems. The new smart grids are capable of not only real-time creation and processing of multiple streams of data to determine optimal power distribution to fulfill real customer needs, but also generating accurate short-term predictions aimed at covering unexpected demand and renewable energy generation peaks. Figure 3.10 shows a depiction of a generic use case for streaming analytics in the energy industry (a typical smart grid application). The goal is to accurately predict electricity demand and production in real time by using streaming data that is coming from smart meters, production system sensors, and meteorological models. The ability to predict near future consumption/production trends and detect anomalies in real time can be used to optimize supply decisions (how much to produce, what sources of production to use, and optimally adjust production capacities) as well as to adjust smart meters to regulate consumption and favorable energy pricing.

Applications of Stream Analytics

Because of its power to create insight instantly, helping decision makers to be on top of events as they unfold and allowing organizations to address issues before they become problems, the use of streaming analytics is on an exponentially increasing trend. The following are some of the application areas that have already benefited from stream analytics.

e-Commerce

Companies like Amazon and eBay (among many others) are trying to make the most out of the data that they collect while a customer is on their Web site. Every page visit, every product looked at, every search conducted, and every click made is recorded and

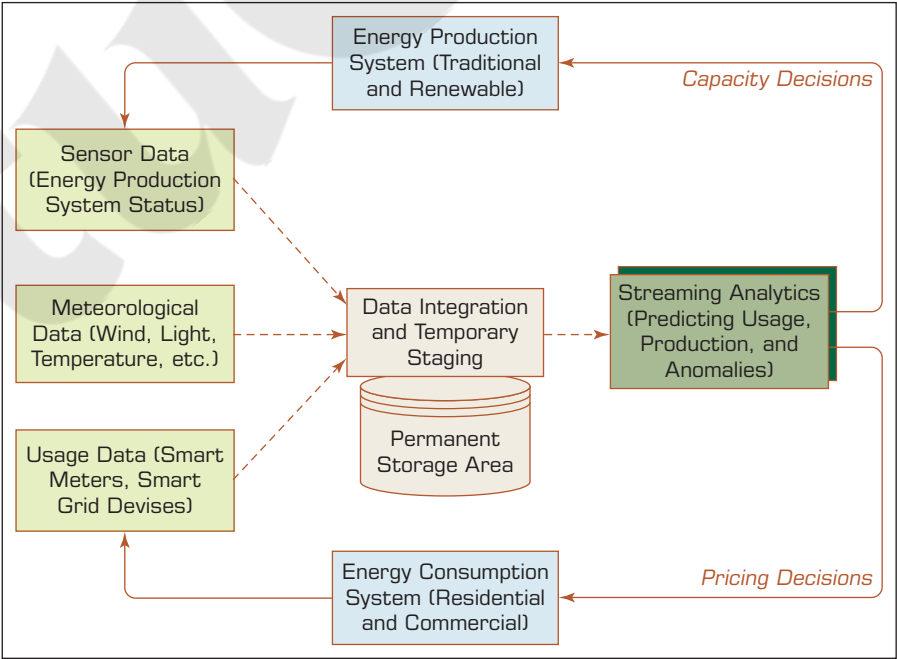


FIGURE 3.10 A Use Case of Streaming Analytics in the Energy Industry.

analyzed to maximize the value gained from a user's visit. If done quickly, analysis of such a stream of data can turn browsers into buyers and buyers into shopaholics. When we visit an e-commerce Web site, even the ones where we are not a member, after a few clicks here and there we start to get very interesting product and bundle price offers. Behind the scenes, advanced analytics are crunching the real-time data coming from our clicks, and the clicks of thousands of others, to "understand" what it is that we are interested in (in some cases, even we do not know that) and make the most of that information by making creative offerings.

Telecommunications

The volume of data that come from call detail records (CDR) for telecommunications companies is astounding. Although this information has been used for billing purposes for quite some time now, there is a wealth of knowledge buried deep inside this Big Data that the telecommunications companies are just now realizing to tap. For instance, CDR data can be analyzed to prevent churn by identifying networks of callers, influencers, leaders, and followers within those networks and proactively acting on this information. As we all know, influencers and leaders have the effect of changing the perception of the followers within their network toward the service provider, either positively or negatively. Using social network analysis techniques, telecommunication companies are identifying the leaders and influencers and their network participants to better manage their customer base. In addition to churn analysis, such information can also be used to recruit new members and maximize the value of the existing members.

Continuous streams of data that come from CDR can be combined with social media data (sentiment analysis) to assess the effectiveness of marketing campaigns. Insight gained from these data streams can be used to rapidly react to adverse effects (which may lead to loss of customers) or boost the impact of positive effects (which may lead to maximizing purchases of existing customers and recruitment of new customers) observed in these campaigns. Furthermore, the process of gaining insight from CDR can be replicated for data networks using Internet protocol detail records. Because most telecommunications companies provide both of these service types, a holistic optimization of all offerings and marketing campaigns could lead to extraordinary market gains.

Law Enforcement and Cybersecurity

Streams of Big Data provide excellent opportunities for improved crime prevention, law enforcement, and enhanced security. They offer unmatched potential when it comes to security applications that can be built in the space, such as real-time situational awareness, multimodal surveillance, cyber-security detection, legal wiretapping, video surveillance, and face recognition (Zikopoulos et al., 2013). As an application of information assurance, enterprises can use streaming analytics to detect and prevent network intrusions, cyberattacks, and malicious activities by streaming and analyzing network logs and other Internet activity monitoring resources.

Power Industry

Because of the increasing use of smart meters, the amount of real-time data collected by power utilities is increasing exponentially. Moving from once a month to every 15 minutes (or more frequently), meter reading accumulates large quantities of invaluable data for power utilities. These smart meters and other sensors placed all around the power grid are sending information back to the control centers to be analyzed in real time. Such analyses help utility companies to optimize their supply chain decisions (e.g., capacity adjustments, distribution network options, real-time buying or selling) based on the

up-to-the-minute consumer usage and demand patterns. In addition, utility companies can integrate weather and other natural conditions data into their analytics to optimize power generation from alternative sources (e.g., wind, solar) and to better forecast energy demand on different geographic granulations. Similar benefits also apply to other utilities such as water and natural gas.

Financial Services

Financial service companies are among the prime examples where analysis of Big Data streams can provide faster and better decisions, competitive advantage, and regulatory oversight. The ability to analyze fast-paced, high-volumes of trading data at very low latency across markets and countries offers a tremendous advantage to making the split-second buy/sell decisions that potentially translate into big financial gains. In addition to optimal buy/sell decisions, stream analytics can also help financial service companies in real-time trade monitoring to detect fraud and other illegal activities.

Health Sciences

Modern-era medical devices (e.g., electrocardiograms and equipment that measure blood pressure, blood oxygen level, blood sugar level, and body temperature) are capable of producing invaluable streaming diagnostic/sensory data at a very fast rate. Harnessing this data and analyzing it in real time offers benefits—the kind that we often call “life and death”—unlike any other field. In addition to helping healthcare companies become more effective and efficient (and hence more competitive and profitable), stream analytics is also improving patient conditions and saving lives.

Many hospital systems all around the world are developing care infrastructures and health systems that are futuristic. These systems aim to take full advantage of what the technology has to offer, and more. Using hardware devices that generate high-resolution data at a very rapid rate, coupled with super-fast computers that can synergistically analyze multiple streams of data, increases the chances of keeping patients safe by quickly detecting anomalies. These systems are meant to help human decision makers make faster and better decisions by being exposed to a multitude of information as soon as it becomes available.

Government

Governments around the world are trying to find ways to be more efficient (via optimal use of limited resources) and effective (providing the services that people need and want). As the practices for e-government become mainstream, coupled with widespread use and access to social media, very large quantities of data (both structured and unstructured) are at the disposal of government agencies. Proper and timely use of these Big Data streams differentiates proactive and highly efficient agencies from the ones who are still using traditional methods to react to situations as they unfold. Another way in which government agencies can leverage real-time analytics capabilities is to manage natural disasters such as snowstorms, hurricanes, tornadoes, and wildfires through a surveillance of streaming data coming from radar, sensors, and other smart detection devices. They can also use similar approaches to monitor water quality, air quality, and consumption patterns and detect anomalies before they become significant problems. Another area where government agencies use stream analytics is in traffic management in congested cities. By using the data coming from traffic flow cameras, GPS data coming from commercial vehicles, and traffic sensors embedded in roadways, agencies are able to change traffic light sequences and traffic flow lanes to ease the pain caused by traffic congestion problems.

SECTION 3.8 REVIEW QUESTIONS

1. What is stream analytics? How does it differ from regular analytics?
2. What are the motivations for stream analytics?
3. What are the most fruitful industries for stream analytics?
4. How can stream analytics be used in e-commerce?
5. In addition to what is listed in this section, can you think of other industries and/or application areas where stream analytics can be used?
6. Compared to regular analytics, do you think stream analytics will have more (or less) use cases in the era of Big Data analytics? Why?

3.9 STATISTICAL MODELING FOR BUSINESS ANALYTICS

Because of the increasing popularity of business analytics, the traditional statistical methods and underlying techniques are also regaining their attractiveness as enabling tools to support evidence-based managerial decision making. Not only are they regaining attention and admiration, but this time around, they are attracting business users in addition to statisticians and analytics professionals.

Statistics (statistical methods and underlying techniques) is usually considered as part of descriptive analytics (see Figure 3.11). Some of the statistical methods can also be considered as part of predictive analytics such as discriminant analysis, multiple regression, logistic regression, and k-means clustering. As shown in Figure 3.11, descriptive analytics has two main branches: statistics and **online analytics processing (OLAP)**. OLAP is the term used for analyzing, characterizing, and summarizing structured data stored in organizational databases (often stored in a data warehouse or in a data mart) using cubes (i.e., multidimensional data structures that are created to extract a subset of data values to answer a specific business question). The OLAP branch of descriptive analytics has also been called business intelligence. Statistics, on the other hand, helps to characterize the data either one variable at a time or multivariables all together, using either descriptive or inferential methods.

Statistics—a collection of mathematical techniques to characterize and interpret data—has been around for a very long time. Many methods and techniques have been developed to address the needs of the end users and the unique characteristics of the

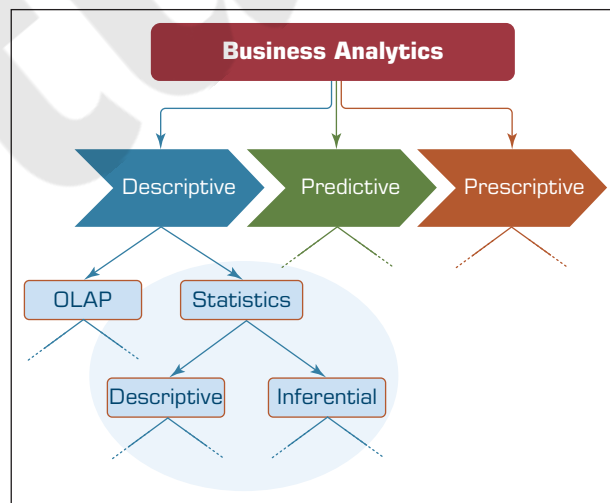


FIGURE 3.11 Relationship between Statistics and Descriptive Analytics.

data being analyzed. Generally speaking, at the highest level, statistical methods can be classified as either descriptive or inferential. The main difference between descriptive and inferential statistics is the data used in these methods—whereas **descriptive statistics** is all about describing the sample data on hand, and **inferential statistics** is about drawing inferences or conclusions about the characteristics of the population. In this section we will briefly describe descriptive statistics (because of the fact that it lays the foundation for, and is the integral part of, descriptive analytics), and in the following section we will cover regression (both linear and logistic regression) as part of inferential statistics.

Descriptive Statistics for Descriptive Analytics

Descriptive statistics, as the name implies, describes the basic characteristics of the data at hand, often one variable at a time. Using formulas and numerical aggregations, descriptive statistics summarizes the data in such a way that often meaningful and easily understandable patterns emerge from the study. Although it is very useful in data analytics and very popular among the statistical methods, descriptive statistics does not allow making conclusions (or inferences) beyond the sample of the data being analyzed. That is, it is simply a nice way to characterize and describe the data on hand, without making conclusions (inferences or extrapolations) regarding the population of related hypotheses we might have in mind.

In business analytics, descriptive statistics plays a critical role—it allows us to understand and explain/present our data in a meaningful manner using aggregated numbers, data tables, or charts/graphs. In essence, descriptive statistics helps us convert our numbers and symbols into meaningful representations for anyone to understand and use. Such an understanding not only helps business users in their decision-making processes, but also helps analytics professionals and data scientists to characterize and validate the data for other more sophisticated analytics tasks. Descriptive statistics allows analysts to identify data concentration, unusually large or small values (i.e., outliers), and unexpectedly distributed data values for numeric variables. Therefore, the methods in descriptive statistics can be classified as either measures for central tendency or measures of dispersion. In the following section, we will use a simple description and mathematical formulation/representation of these measures. In mathematical representation, we will use $x_1, x_2, \dots, x_n$ to represent individual values (observations) of the variable (measure) that we are interested in characterizing.

Measures of Centrality Tendency (May Also Be Called Measures of Location or Centrality)

Measures of centrality are the mathematical methods by which we estimate or describe central positioning of a given variable of interest. A measure of central tendency is a single numerical value that aims to describe a set of data by simply identifying or estimating the central position within the data. The mean (often called the arithmetic mean or the simple average) is the most commonly used measure of central tendency. In addition to mean, you could also see median or mode being used to describe the centrality of a given variable. Although, the mean, median, and mode are all valid measures of central tendency, under different circumstances, one of these measures of centrality becomes more appropriate than the others. What follows are short descriptions of these measures, including how to calculate them mathematically and pointers on the circumstances in which they are the most appropriate measure to use.

Arithmetic Mean

The **arithmetic mean** (or simply *mean* or *average*) is the sum of all the values/observations divided by the number of observations in the data set. It is by far the most popular and most commonly used measure of central tendency. It is used with continuous or

discrete numeric data. For a given variable x , if we happen to have n values/observations ($x_1, x_2, \dots, x_n$), we can write the arithmetic mean of the data sample ($\bar{x}$, pronounced as x-bar) as follows:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

or

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

The mean has several unique characteristics. For instance, the sum of the absolute deviations (differences between the mean and the observations) above the mean is the same as the sum of the deviations below the mean, balancing the values on either side of it. That said, it does not suggest, however, that half the observations are above and the other half are below the mean (a common misconception among those who do not know basic statistics). Also, the mean is unique for every data set and is meaningful and calculable for both interval- and ratio-type numeric data. One major downside is that the mean can be affected by outliers (observations that are considerably larger or smaller than the rest of the data points). Outliers can pull the mean toward their direction and, hence, bias the centrality representation. Therefore, if there are outliers or if the data is erratically dispersed and skewed, one should either avoid using mean as the measure of centrality or augment it with other central tendency measures, such as median and mode.

Median

The **median** is the measure of center value in a given data set. It is the number in the middle of a given set of data that has been arranged/sorted in order of magnitude (either ascending or descending). If the number of observation is an odd number, identifying the median is very easy—just sort the observations based on their values and pick the value right in the middle. If the number of observations is an even number, then identify the two middle values, and then take the simple average of these two values. The median is meaningful and calculable for ratio, interval, and ordinal data types. Once determined, half the data points in the data are above and the other half are below the median. In contrary to the mean, the median is not affected by outliers or skewed data.

Mode

The **mode** is the observation that occurs most frequently (the most frequent value in our data set). On a histogram it represents the highest bar in a bar chart, and hence it may be considered as being the most popular option/value. The mode is most useful for data sets that contain a relatively small number of unique values. That is, it may be useless if the data have too many unique values (as is the case in many engineering measurements that capture high precision with a large number of decimal places), rendering each value having either one or a very small number representing its frequency. Although it is a useful measure (especially for nominal data), mode is not a very good representation of centrality, and therefore it should not be used as the only measure of central tendency for a given data set.

In summary, which central tendency measure is the best? Although there is not a clear answer to this question, here are a few hints—use the mean when the data is not prone to outliers and there is no significant level of skewness; use the median when the data has outliers and/or it is ordinal in nature; use the mode when the data is nominal. Perhaps the best practice is to use all three together so that the central tendency of the data set can be captured and represented from three perspectives. Mostly because

“average” is a very familiar and highly used concept to everyone in regular daily activities, managers (as well as some scientists and journalists) often use the centrality measures (especially mean) inappropriately when other statistical information should be considered along with the centrality. It is a better practice to present descriptive statistics as a package—a combination of centrality and dispersion measures—as opposed to a single measure like mean.

Measures of Dispersion

Measures of **dispersion** (it may also be called measures of spread or decentrality) are the mathematical methods used to estimate or describe the degree of variation in a given variable of interest. They are a representation of the numerical spread (compactness or lack thereof) of a given data set. To describe this dispersion, a number of statistical measures are developed; the most notable ones are range, variance, and standard deviation (and also quartiles and absolute deviation). One of the main reasons why the measures of dispersion/spread of data values are important is the fact that it gives us a framework within which we can judge the central tendency—gives us the indication of how well the mean (or other centrality measures) represents the sample data. If the dispersion of values in the data set is large, the mean is not deemed to be a very good representation of the data. This is because a large dispersion measure indicates large differences between individual scores. Also, in research, it is often perceived as a positive sign to see a small variation within each data sample, as it may indicate homogeneity, similarity, and robustness within the collected data.

Range

The **range** is perhaps the simplest measure of dispersion. It is the difference between the largest and the smallest values in a given data set (i.e., variables). So, we calculate range by simply identifying the smallest value in the data set (minimum), identifying the largest value in the data set (maximum), and calculating the difference between them (range = maximum – minimum).

Variance

A more comprehensive and sophisticated measure of dispersion is the **variance**. It is a method used to calculate the deviation of all data points in a given data set from the mean. The larger the variance, the more the data are spread out from the mean and the more variability one can observe in the data sample. To prevent the offsetting of negative and positive differences, the variance takes into account the square of the distances from the mean. The formula for a data sample can be written as

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

where n is the number of samples, $\bar{x}$ is the mean of the sample, and x_i is the i^{th} value in the data set. The larger values of variance indicate more dispersion, whereas smaller values indicate compression in the overall data set. Because the differences are squared, larger deviations from the mean contribute significantly to the value of variance. Again, because the differences are squared, the numbers that represent deviation/variance become somewhat meaningless (as opposed to a dollar difference, herein you are given a squared dollar difference). Therefore, instead of variance, in many business applications we use a more meaningful dispersion measure, called standard deviation.

Standard Deviation

The **standard deviation** is also a measure of the spread of values within a set of data. The standard deviation is calculated by simply taking the square root of the variations. The following formula shows the calculation of standard deviation from a given sample of data points.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Mean Absolute Deviation

In addition to variance and standard deviation, sometimes we also use **mean absolute deviation** to measure dispersion in a data set. It is a simpler way to calculate the overall deviation from the mean. Specifically, it is calculated by measuring the absolute values of the differences between each data point and the mean and summing them. It provides a measure of spread without being specific about the data point being lower or higher than the mean. The following formula shows the calculation of the mean absolute deviation:

$$MAD = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

Quartiles and Interquartile Range

Quartiles help us identify spread within a subset of the data. A **quartile** is a quarter of the number of data points given in a data set. Quartiles are determined by first sorting the data and then splitting the sorted data into four disjoint smaller data sets. Quartiles are a useful measure of dispersion because they are much less affected by outliers or a skewness in the data set than the equivalent measures in the whole data set. Quartiles are often reported along with the median as the best choice of measure of dispersion and central tendency, respectively, when dealing with skewed and/or data with outliers. A common way of expressing quartiles is as an interquartile range, which describes the difference between the third quartile (Q3) and the first quartile (Q1), telling us about the range of the middle half of the scores in the distribution. The quartile-driven descriptive measures (both centrality and dispersion) are best explained with a popular plot called a box plot (or box-and-whiskers plot).

Box-and-Whiskers Plot

The **box-and-whiskers plot** (or simply a **box plot**) is a graphical illustration of several descriptive statistics about a given data set. They can be either horizontal or vertical, but vertical is the most common representation, especially in modern-day analytics software products. It is known to be first created and presented by John W. Tukey in 1969. Box plot is often used to illustrate both centrality and dispersion of a given data set (i.e., the distribution of the sample data) in an easy-to-understand graphical notation. Figure 3.12 shows a couple of box plots side by side, sharing the same y -axis. As shown therein, a single chart can have one or more box plots for visual comparison purposes. In such cases, the y -axis would be the common measure of magnitude (the numerical value of the variable), with the x -axis showing different classes/subsets such as different time dimensions (e.g., descriptive statistics for annual Medicare expenses in 2015 versus 2016) or different categories (e.g., descriptive statistics for marketing expenses versus total sales).

Although, historically speaking, the box plot was not used widely and often enough (especially in areas outside of statistics), with the emerging popularity of business

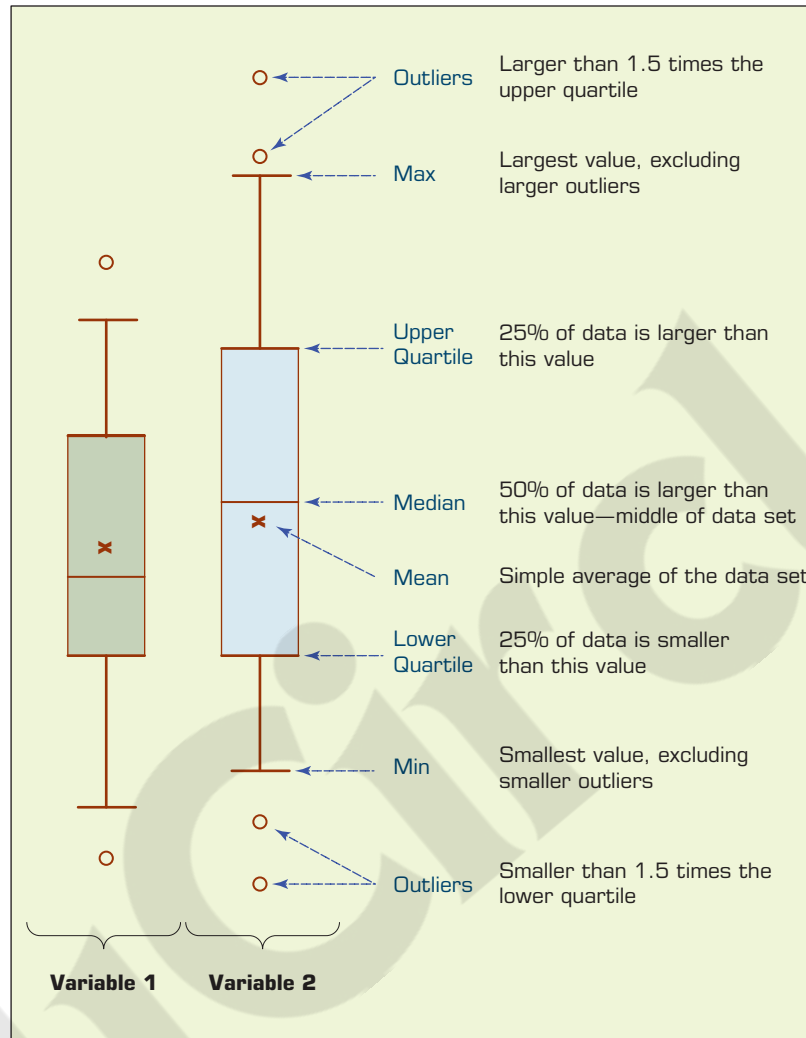


FIGURE 3.12 Understanding the Specifics about Box-and-Whiskers Plots.

analytics, it is gaining fame in less-technical areas of the business world. Its information richness and ease of understanding are largely to credit for its recent popularity.

The box plot shows the **centrality** (median, and sometimes also mean) as well as the dispersion (the density of the data within the middle half—drawn as a box between the first and third quartile), the minimum and maximum ranges (shown as extended lines from the box, looking like whiskers, that are calculated as 1.5 times the upper or lower end of the quartile box) along with the outliers that are larger than the limits of the whiskers. A box plot also shows whether the data is symmetrically distributed with respect to the mean, or it sways one way or another. The relative position of the median versus mean and the lengths of the whiskers on both side of the box give a good indication of the potential skewness in the data.

The Shape of a Distribution

Although not as common as the centrality and dispersion, the shape of the data distribution is also a useful measure for the descriptive statistics. Before delving into the shape of the distribution we first need to define the distribution itself. Simply put, distribution

is the frequency of data points counted and plotted over a small number of class labels or numerical ranges (i.e., bins). In a graphical illustration of distribution, the y -axis shows the frequency (count or %), and the x -axis shows the individual classes or bins in a rank-ordered fashion. A very well-known distribution is called normal distribution, which is perfectly symmetric on both sides of the mean and has several well-founded mathematical properties that make it a very useful statistical tool for research and practice. As the dispersion of a data set increases, so does the standard deviation, and the shape of the distribution looks wider. A graphic illustration of the relationship between dispersion and distribution shape (in the context of normal distribution) is shown in Figure 3.13.

There are two commonly used measures to calculate the shape characteristics of a distribution: skewness and kurtosis. A **histogram** (i.e., frequency plot) is often used to visually illustrate both skewness and kurtosis.

Skewness is a measure of asymmetry (sway) in a distribution of the data that portrays a unimodal (i.e., only one peak exists in the distribution) structure of the data. Because normal distribution is a perfectly symmetric unimodal distribution, it does not have skewness, that is, its skewness measure (i.e., the value of the coefficient of skewness) is equal to zero. The skewness measure/value can be either positive or negative. If the distribution sways left (i.e., the peak is on the left and the long tail is on the right side, and the mean is greater than median), then it produces a positive skewness measure, and if the distribution sways right (i.e., the peak is on the right and the long tail is

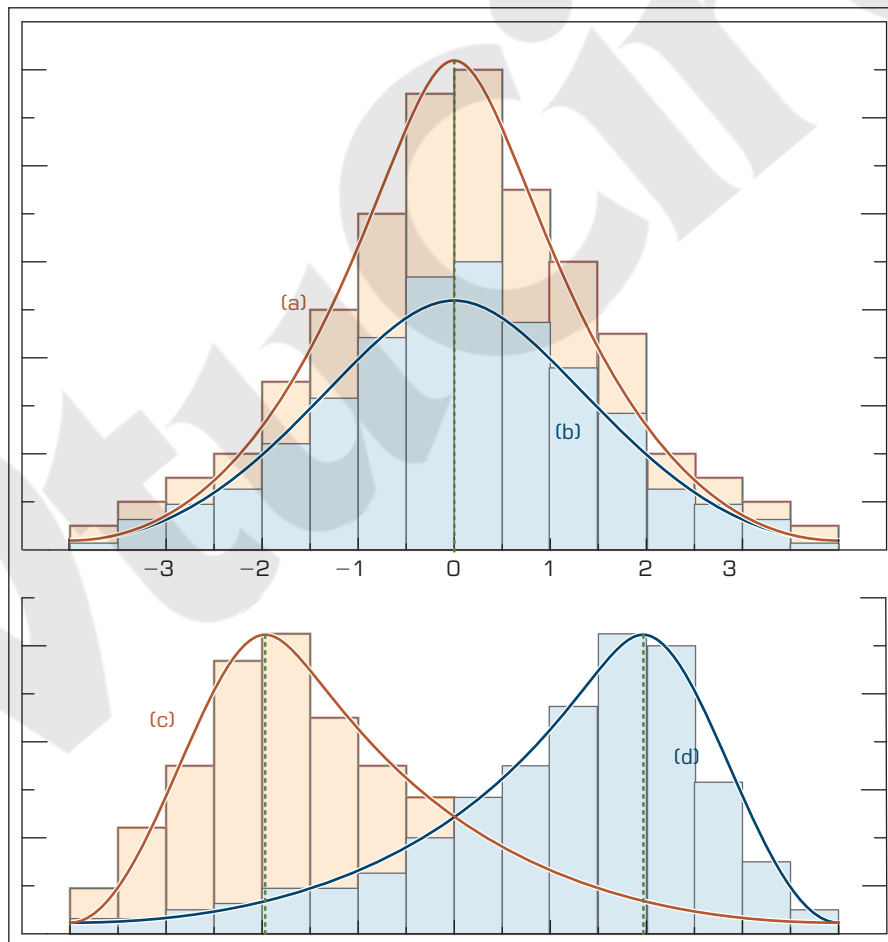


FIGURE 3.13 Relationship between Dispersion and Shape Properties.

on the left side, and the mean is smaller than median), then it produces a negative skewness measure. In Figure 3.13, (c) represents a positively skewed distribution, whereas (d) represents a negatively skewed distribution. In the same figure, both (a) and (b) represent perfect symmetry and hence zero measure for skewness.

$$\text{Skewness} = S = \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{(n-1)s^3}$$

where s is the standard deviation and n is the number of samples.

Kurtosis is another measure to use in characterizing the shape of a unimodal distribution. As opposed to the sway in shape, kurtosis focuses on characterizing the peakiness/tallness nature of the distribution. Specifically, kurtosis measures the degree to which a distribution is more or less peaked than a normal distribution. Whereas a positive kurtosis indicates a relatively peaked/tall distribution, a negative kurtosis indicates a relatively flat/short distribution. As a reference point, a normal distribution has a kurtosis of 3. The formula for kurtosis can be written as

$$\text{Kurtosis} = K = \frac{\sum_{i=1}^n (x_i - \bar{x})^4}{ns^4} - 3$$

Descriptive statistics (as well as inferential statistics) can easily be calculated using commercial statistical software packages (e.g., SAS, SPSS, Minitab, JMP, Statistica) or free/open-source tools (e.g., R, Python, KNIME). Perhaps the most convenient way to calculate descriptive and some of the inferential statistics is to use Excel. Technology Insight 3.3 describes in detail how to use Microsoft Excel to calculate descriptive statistics.

TECHNOLOGY INSIGHT 3.3 How to Calculate Descriptive Statistics in Microsoft Excel

Excel, arguably the most popular data analysis tool in the world, can easily be used for descriptive statistics. Although the base configuration of Excel does not seem to have the statistics function readily available for end users, those functions come with the installation and can be activated (turned on) with only a few mouse clicks. Figure 3.14 shows how these statistics functions (as part of the Analysis ToolPak) can be activated in Microsoft Excel.

Once activated, the *Analysis ToolPak* will appear in the *Data* menu option under the name of *Data Analysis*. When you click on Data Analysis in the Analysis group under the Data tab in the Excel menu bar, you will see Descriptive Statistics as one of the options within the list of data analysis tools (see Figure 3.15, steps [1, 2]); click on OK, and the Descriptive Statistics dialog box will appear (see middle of Figure 3.15). In this dialog box you need to enter the range of the data, which can be one or more numerical columns, along with the preference check boxes, and click OK (see Figure 3.15, steps [3, 4]). If the selection includes more than one numeric column, the tool treats each column as a separate data set and provides descriptive statistics for each column separately.

As a simple example, we selected two columns (labeled as Expense and Demand) and executed the Descriptive Statistics option. The bottom section of Figure 3.15 shows the output created by Excel. As can be seen, Excel produced all the descriptive statistics that are covered in the previous section and added a few more to the list. In Excel, it is also very easy (a few mouse clicks) to create a box-and-whiskers plot. Figure 3.16 shows the simple three-step process of creating a box-and-whiskers plot in Excel.

Although this is a very useful tool in Excel, one should be aware of an important point related to the results generated by Analysis ToolPak, which have a different behavior than other ordinary Excel functions: Although Excel functions dynamically change as the underlying

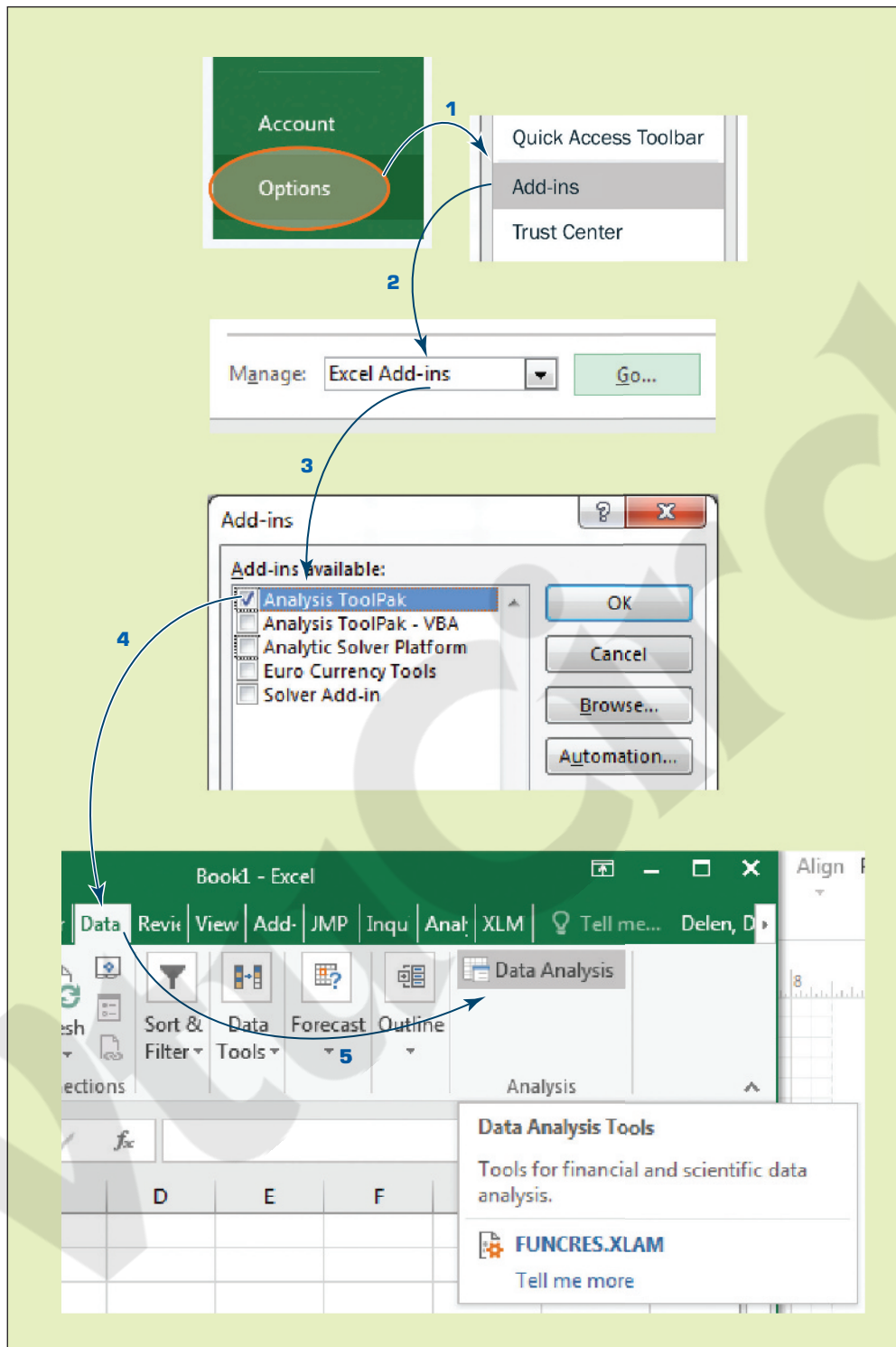
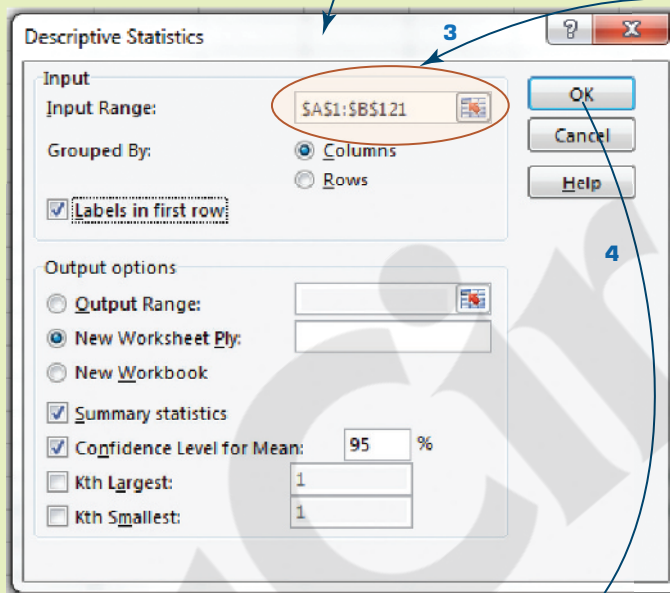
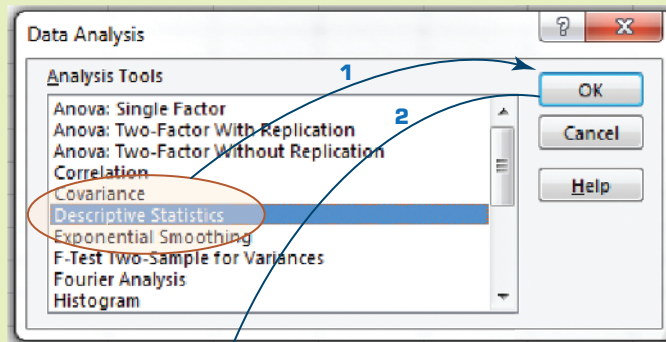


FIGURE 3.14 Activating Data Analysis Functions in Excel.



	A	B
1	Expense	Demand
2	8553	14824
3	9107	14567
4	8137	14509
5	6999	12395
6	6009	11031
7	8140	12945
8	5811	11558
9	7787	14018
10	6032	11828
11	7146	11359
12	7379	11663
13	7988	13167
14	6236	11943
15	6967	11524
16	8765	13722
17	6865	11095

	A	B	C	D
1	Expense		Demand	
2				
3	Mean	7595.808333	Mean	12976.53333
4	Standard Error	78.07347263	Standard Error	109.7357916
5	Median	7679	Median	12945
6	Mode	8140	Mode	14567
7	Standard Deviation	855.252042	Standard Deviation	1202.095369
8	Sample Variance	731456.0554	Sample Variance	1445033.276
9	Kurtosis	-0.82138214	Kurtosis	-1.290541404
10	Skewness	-0.235479407	Skewness	-0.002281156
11	Range	3526	Range	3944
12	Minimum	5811	Minimum	11019
13	Maximum	9337	Maximum	14963
14	Sum	911497	Sum	1557184
15	Count	120	Count	120
16	Confidence Level(95.0%)	154.5932735	Confidence Level(95.0%)	217.2878275

FIGURE 3.15 Obtaining Descriptive Statistics in Excel.

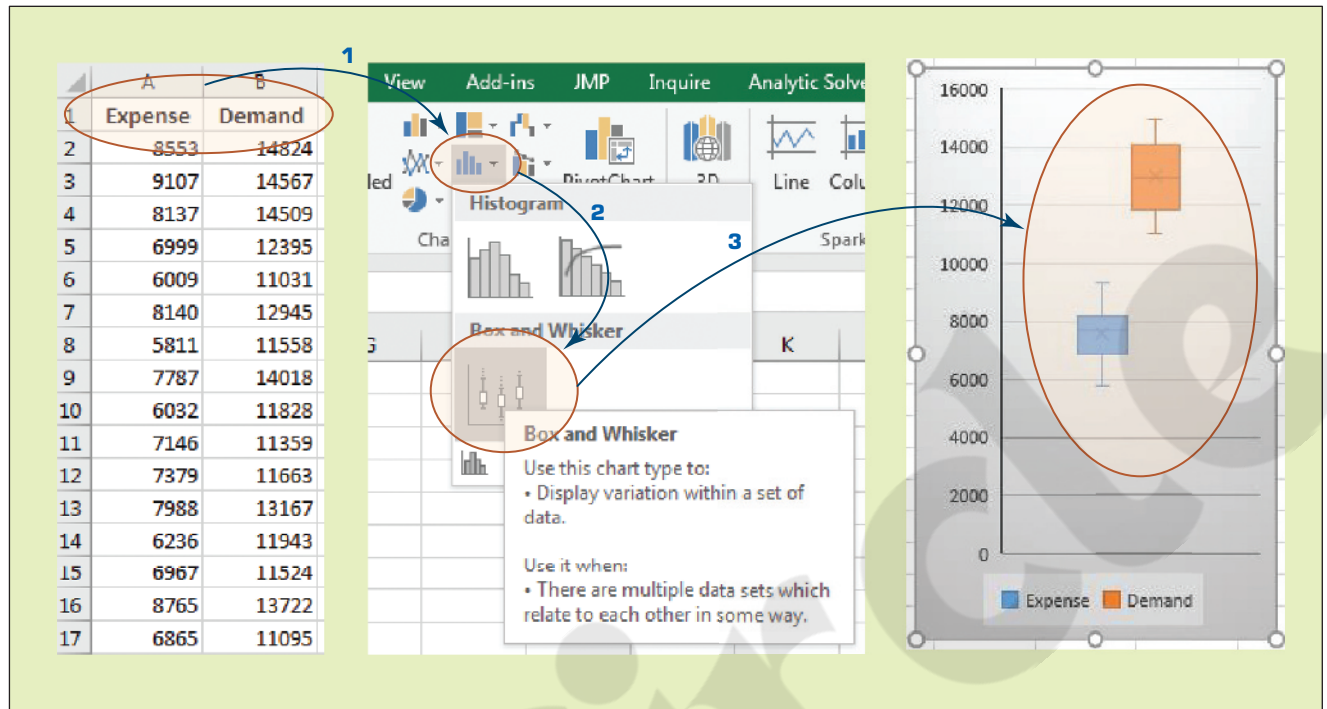


FIGURE 3.16 Creating a Box-and-Whiskers Plot in Excel.

data in the spreadsheet are changed, the results generated by the Analysis ToolPak do not. For example, if you change the values in either or both of these columns, the Descriptive Statistics results produced by the Analysis ToolPak will stay the same. However, the same is not true for ordinary Excel functions. If you were to calculate the mean value of a given column (using `=AVERAGE(A1:A121)`), and then change the values within the data range, the mean value would automatically change. In summary, the results produced by Analysis ToolPak do not have a dynamic link to the underlying data, and if the data changes, the analysis needs to be redone using the dialog box.

Successful applications of data analytics cover a wide range of business and organizational settings, addressing the problems once thought unsolvable. Analytics in Action 3.2 is an excellent illustration of those success stories where a small municipality administration adopts a data analytics approach to intelligently detect and solve problems by continuously analyzing demand and consumption patterns.

ANALYTICS IN ACTION 3.2 Town of Cary Uses Analytics to Analyze Data from Sensors, Assess Demand, and Detect Problems

A leaky faucet. A malfunctioning dishwasher. A cracked sprinkler head. These are more than just a headache for a home owner or business to fix. They can be costly, unpredictable, and, unfortunately, hard to pinpoint. Through a combination of wireless water meters and a data-analytics-driven, customer-accessible portal, the Town of Cary, North Carolina, is making it much easier to find and fix water loss issues. In the process, the town has gained a big-picture view of water usage critical to planning future water plant expansions and promoting targeted conservation efforts.

When the Town of Cary installed the wireless meters for 60,000 customers in 2010, it knew the new technology wouldn't just save money by eliminating manual monthly readings; the town also realized it would get more accurate and timely information about water consumption. The Aquastar wireless system reads meters once an hour—that's 8,760 data points per customer each year instead of 12 monthly readings. The data had tremendous potential, if it could be easily consumed.

"Monthly readings are like having a gallon of water's worth of data. Hourly meter readings are more like an Olympic-size pool of data," says Karen Mills, Finance Director for the Town of Cary. "SAS helps us manage the volume of that data nicely." In fact, the solution enables the town to analyze a half-billion data points on water usage and make them available, and easily consumable, to all customers.

The ability to visually look at data by household or commercial customer, by the hour, has led to some very practical applications:

- The town can notify customers of potential leaks within days.
- Customers can set alerts that notify them within hours if there is a spike in water usage.
- Customers can track their water usage online, helping them to be more proactive in conserving water.

Through the online portal, one business in the Town of Cary saw a spike in water consumption on weekends, when employees are away. This seemed odd, and the unusual reading helped the company learn that a commercial dishwasher was malfunctioning, running continuously over weekends. Without the wireless water-meter data and the customer-accessible portal, this problem could have gone unnoticed, continuing to waste water and money.

The town has a much more accurate picture of daily water usage per person, critical for planning future water plant expansions. Perhaps the most interesting perk is that the town was able to verify a hunch that has far-reaching cost ramifications: Cary residents are very economical in their use of water. "We calculate that with modern high-efficiency appliances, indoor water use could be as low as 35 gallons per person per day. Cary residents average 45 gallons, which is still phenomenally low," explains town Water Resource Manager Leila Goodwin. Why is this important? The town was spending money to encourage water efficiency—rebates on low-flow toilets or discounts on rain barrels. Now it can take a more targeted approach, helping specific consumers understand and manage both their indoor and outdoor water use.

SAS was critical not just for enabling residents to understand their water use, but also in working behind the scenes to link two disparate databases. "We have a billing database and the meter-reading database. We needed to bring that together and make it presentable," Mills says.

The town estimates that by just removing the need for manual readings, the Aquastar system will save more than \$10 million above the cost of the project. But the analytics component could provide even bigger savings. Already both the town and individual citizens have saved money by catching water leaks early. As the Town of Cary continues to plan its future infrastructure needs, having accurate information on water usage will help it invest in the right amount of infrastructure at the right time. In addition, understanding water usage will help the town if it experiences something detrimental like a drought.

"We went through a drought in 2007," says Goodwin. "If we go through another, we have a plan in place to use Aquastar data to see exactly how much water we are using on a day-by-day basis and communicate with customers. We can show 'here's what's happening, and here is how much you can use because our supply is low.' Hopefully, we'll never have to use it, but we're prepared."

Source: "Municipality puts wireless water meter-reading data to work (SAS® Analytics) – The Town of Cary, North Carolina uses SAS Analytics to analyze data from wireless water meters, assess demand, detect problems and engage customers" (accessed May 2022). Copyright © 2022 SAS Institute Inc., Cary, NC, USA. Reprinted with permission. All rights reserved.

SECTION 3.9 REVIEW QUESTIONS

1. What is the relationship between statistics and business analytics?
2. What are the main differences between descriptive and inferential statistics?
3. List and briefly define the central tendency measures of descriptive statistics.
4. List and briefly define the dispersion measures of descriptive statistics.
5. What is a box-and-whiskers plot? What types of statistical information does it represent?
6. What are the two most commonly used shape characteristics to describe a data distribution?

3.10 REGRESSION MODELING FOR INFERENCE STATISTICS

Regression, especially linear regression, is perhaps the most widely known and used analytics technique in statistics. Historically speaking, the roots of regression date back to the 1920s and 1930s, to the earlier work on inherited characteristics of sweet peas by Sir Francis Galton and subsequently by Karl Pearson. Since then regression has become the statistical technique for characterization of relationships between explanatory (input) variable(s) and response (output) variable(s).

As popular as it is, essentially, regression is a relatively simple statistical technique to model the dependence of a variable (response or output variable) on one (or more) explanatory (input) variables. Once identified, this relationship between the variables can be formally represented as a linear/additive function/equation. As is the case with many other modeling techniques, regression aims to capture the functional relationship between and among the characteristics of the real world and describe this relationship with a mathematical model, which may then be used to discover and understand the complexities of reality—explore and explain relationships or forecast future occurrences.

Regression can be used for one of two purposes: hypothesis testing—investigating potential relationships between different variables, and prediction/forecasting—estimating values of a response variables based on one or more explanatory variables. These two uses are not mutually exclusive. The explanatory power of regression is also the foundation of its prediction ability. In hypothesis testing (theory building), regression analysis can reveal the existence/strength and the directions of relationships between a number of explanatory variables (often represented with x_i) and the response variable (often represented with y). In prediction, regression identifies additive mathematical relationships (in the form of an equation) between one or more explanatory variables and a response variable. Once determined, this equation can be used to forecast the values of the response variable for a given set of values of the explanatory variables.

Correlation versus Regression Because regression analysis originated from correlation studies, and because both methods attempt to describe the association between two (or more) variables, these two terms are often confused by professionals and even by scientists. **Correlation** makes no a priori assumption of whether one variable is dependent on the other(s) and is not concerned with the relationship between variables; instead it gives an estimate on the degree of association between the variables. On the other hand, regression attempts to describe the dependence of a response variable on one (or more) explanatory variables where it implicitly assumes that there is a one-way causal effect from the explanatory variable(s) to the response variable, regardless of whether the path of effect is direct or indirect. Also, although correlation is interested in the low-level relationships between two variables, regression is concerned with the relationships between all explanatory variables and the response variable.

Simple versus Multiple Regression If the regression equation is built between one response variable and one explanatory variable, then it is called simple regression. For instance, the regression equation built to predict/explain the relationship between a height of a person (explanatory variable) and the weight of a person (response variable) is a good example of simple regression. Multiple regression is the extension of simple regression where the explanatory variables are more than one. For instance, in the previous example, if we were to include not only the height of the person but also other personal characteristics (e.g., BMI, gender, ethnicity) to predict the weight of a person, then we would be performing multiple regression analysis. In both cases, the relationship between the response variable and the explanatory variable(s) are linear and additive in nature. If the relationships are not linear, then we may want to use one of many other nonlinear regression methods to better capture the relationships between the input and output variables.

How Do We Develop the Linear Regression Model?

To understand the relationship between two variables, the simplest and most intuitive thing that one can do is to create a scatter plot, where the y -axis represents the values of the response variable, and the x -axis represents the values of the explanatory variable (see Figure 3.17). A **scatter plot** would show the changes in the response variable as a function of the changes in the explanatory variable. In the case shown in Figure 3.17, there seems to be a positive relationship between the two; as the explanatory variable values increase, so does the response variable.

Simple regression analysis aims to find a mathematical representation of the relationship between two variables. In reality, it tries to find the signature (i.e., algebraic representation) of a straight line passing through right in between the plotted dots (representing the observation/historical data) in such a way that the distance between the dots and the line is minimized (the predicted values on the theoretical regression line). Even

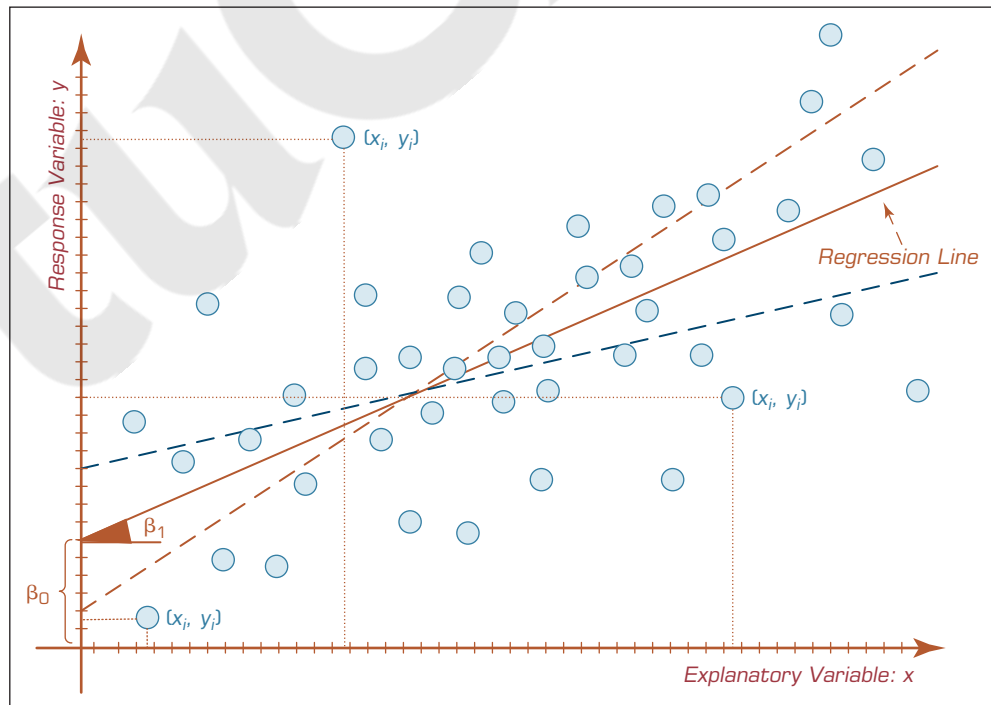


FIGURE 3.17 A Scatter Plot and a Linear Regression Line.

though there are several methods/algorithms proposed to identify the regression line, the one that is most commonly used is called the **ordinary least squares (OLS)** method. The OLS method aims to minimize the sum of squared residuals (squared vertical distances between the observation and the regression point) and leads to a mathematical expression for the estimated value of the regression line (which are known as β parameters). For simple **linear regression**, the aforementioned relationship between the response variable (y) and the explanatory variable(s) (x) can be shown as a simple equation as follows:

$$y = \beta_0 + \beta_1 x$$

In this equation, β_0 is called the intercept and β_1 is called the slope. Once OLS determines the values of these two coefficients, the simple equation can be used to forecast the values of y for given values of x . The sign and the value of β_1 also reveal the direction and the strengths of relationship between the two variables.

If the model is of a multiple linear regression type, then there would be more coefficients to be determined, one for each additional explanatory variable. As the following formula shows, the additional explanatory variable would be multiplied with the new β_i coefficients and summed together to establish a linear additive representation of the response variable.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_n x_n$$

How Do We Know If the Model Is Good Enough?

Because of a variety of reasons, sometimes models as representations of the reality do not prove to be good. Regardless of the number of explanatory variables included, there is always a possibility of not having a good model, and therefore the linear regression model needs to be assessed for its fit (the degree at which it represents the response variable). In the simplest sense, a well-fitting regression model results in predicted values close to the observed data values. For the numerical assessment, three statistical measures are often used in evaluating the fit of a regression model. R^2 (R -squared), the overall F-test, and the root mean square error (RMSE). All three of these measures are based on the sums of the square errors (how far the data are from the mean and how far the data are from the model's predicted values). Different combinations of these two values provide different information about how the regression model compares to the mean model.

Of the three, R^2 has the most useful and understandable meaning because of its intuitive scale. The value of R^2 ranges from zero to one (corresponding to the amount of variability explained in percentage) with zero indicating that the relationship and the prediction power of the proposed model is not good, and one indicating that the proposed model is a perfect fit that produces exact predictions (which is almost never the case). The good R^2 values would usually come close to one, and the closeness is a matter of the phenomenon being modeled—whereas an R^2 value of 0.3 for a linear regression model in social sciences can be considered good enough, an R^2 value of 0.7 in engineering may be considered as not a good-enough fit. The improvement in the regression model can be achieved by adding more explanatory variables, taking some of the variables out of the model, or using different data transformation techniques, which would result in comparative increases in an R^2 value. Figure 3.18 shows the process flow of developing regression models. As can be seen in the process flow, the model development task is followed by the model assessment task, where not only is the fit of the model assessed, but because of restrictive assumptions with which the linear models have to comply, also the validity of the model needs to be put under the microscope.

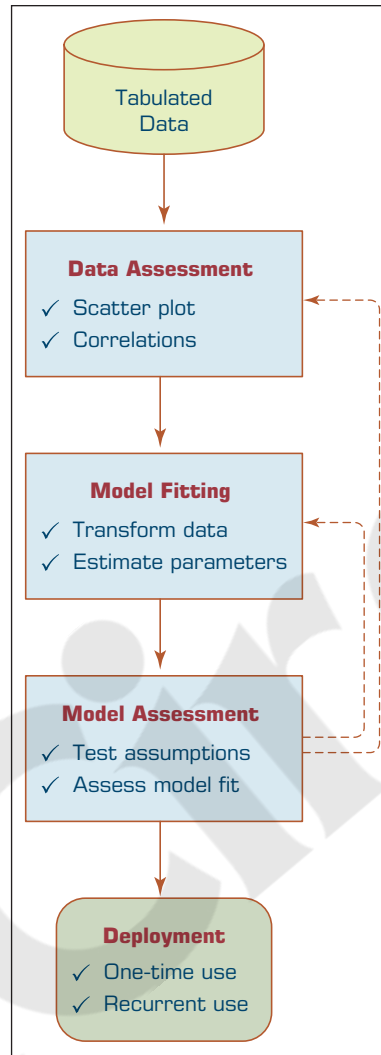


FIGURE 3.18 A Process Flow for Developing Regression Models.

What Are the Most Important Assumptions in Linear Regression?

Even though they are still the choice of many for data analyses (both for explanatory as well as for predictive modeling purposes), linear regression models suffer from several highly restrictive assumptions. The validity of the linear model built depends on its ability to comply with these assumptions. Here are the most commonly pronounced assumptions:

- **Linearity.** This assumption states that the relationship between the response variable and the explanatory variables are linear. That is, the expected value of the response variable is a straight-line function of each explanatory variable, while holding all other explanatory variables fixed. Also, the slope of the line does not depend on the values of the other variables. It also implies that the effects of different explanatory variables on the expected value of the response variable are additive in nature.
- **Independence** (of errors). This assumption states that the errors of the response variable are uncorrelated with each other. This independence of the errors is weaker than actual statistical independence, which is a stronger condition and is often not needed for linear regression analysis.

- **Normality** (of errors). This assumption states that the errors of the response variable are normally distributed. That is, they are supposed to be totally random and should not represent any nonrandom patterns.
- **Constant variance** (of errors). This assumption, also called homoscedasticity, states that the response variables have the same variance in their error, regardless of the values of the explanatory variables. In practice, this assumption is invalid if the response variable varies over a wide enough range/scale.
- **Multicollinearity**. This assumption states that the explanatory variables are not correlated (i.e., do not replicate the same but provide a different perspective of the information needed for the model). Multicollinearity can be triggered by having two or more perfectly correlated explanatory variables presented to the model (e.g., if the same explanatory variable is mistakenly included in the model twice, one with a slight transformation of the same variable). A correlation-based data assessment usually catches this error.

There are statistical techniques developed to identify the violation of these assumptions and techniques to mitigate them. The most important part for a modeler is to be aware of their existence and to put in place the means to assess the models to make sure that the models are compliant with the assumptions they are built on.

Logistic Regression

Logistic regression is a very popular, statistically sound, probability-based classification algorithm that employs supervised **learning**. It was developed in the 1940s as a complement to linear regression and linear discriminant analysis methods. It has been used extensively in numerous disciplines, including the medical and social sciences fields. Logistic regression is similar to linear regression in that it also aims to regress to a mathematical function that explains the relationship between the response variable and the explanatory variables using a sample of past observations (training data). It differs from linear regression with one major point: its output (response variable) is a class as opposed to a numerical variable. That is, whereas linear regression is used to estimate a continuous numerical variable, logistic regression is used to classify a categorical variable. Even though the original form of logistic regression was developed for a binary output variable (e.g., 1/0, yes/no, pass/fail, accept/reject), the present-day modified version is capable of predicting multiclass output variables (i.e., multinomial logistic regression). If there is only one predictor variable and one predicted variable, the method is called simple logistic regression (similar to calling linear regression models with only one independent variable as simple linear regression).

In predictive analytics, logistic regression models are used to develop probabilistic models between one or more explanatory/predictor variables (which may be a mix of both continuous and categorical in nature) and a class/response variable (which may be binomial/binary or multinomial/multiclass). Unlike ordinary linear regression, logistic regression is used for predicting categorical (often binary) outcomes of the response variable—treating the response variable as the outcome of a Bernoulli trial. Therefore, logistic regression takes the natural logarithm of the odds of the response variable to create a continuous criterion as a transformed version of the response variable. Thus the logit transformation is referred to as the link function in logistic regression—even though the response variable in logistic regression is categorical or binomial, the logit is the continuous criterion on which linear regression is conducted. Figure 3.19 shows a logistic regression function where the odds are represented in the x -axis (a linear function of the independent variables), whereas the probabilistic outcome is shown in the y -axis (i.e., response variable values change between 0 and 1).

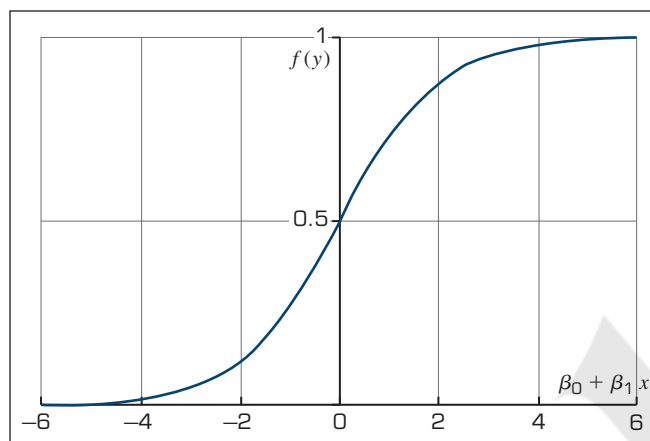


FIGURE 3.19 The Logistic Function.

The logistic function, $f(y)$ in Figure 3.19, is the core of logistic regression, which can only take values between 0 and 1. The following equation is a simple mathematical representation of this function:

$$f(y) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

The logistic regression coefficients (the β s) are usually estimated using the maximum likelihood estimation method. Unlike linear regression with normally distributed residuals, it is not possible to find a closed-form expression for the coefficient values that maximizes the likelihood function, so an iterative process must be used instead. This process begins with a tentative starting solution, then revises the parameters slightly to see if the solution can be improved and repeats this iterative revision until no improvement can be achieved or are very minimal, at which point the process is said to have completed/converged.

Sports analytics—use of data and statistical/analytics techniques to better manage sports teams/organizations—has been gaining tremendous popularity. Use of data-driven analytics techniques has become mainstream for not only professional teams but also college and amateur sports. Analytics in Action 3.3 is an excellent example of how existing and readily available public data sources can be used to predict college football bowl game outcomes using both classification and regression-type prediction models.

ANALYTICS IN ACTION 3.3 Predicting NCAA Bowl Game Outcomes

Predicting the outcome of a college football game (or any sports game, for that matter) is an interesting and challenging problem. Therefore, challenge-seeking researchers from both academics and industry have spent a great deal of effort on forecasting the outcome of sporting events. Large quantities of historic data exist in different media outlets (often publicly available) regarding the structure and outcomes of sporting events in the form of a variety of numerically or symbolically represented factors that are assumed to contribute to those outcomes.

The end-of-season bowl games are very important to colleges both financially (bringing in millions of dollars of additional revenue) as well as reputational—for recruiting quality students and highly regarded high school athletes for their athletic programs (Freeman & Brewer, 2016). Teams that are selected to compete in a given bowl game split a purse, the size of which depends on the specific bowl (some bowls are more prestigious and have higher payouts for the two teams), and therefore securing an invitation to a bowl game is the main goal of any division I-A college football program. The decision makers of the bowl games are given the authority



Source: bestv/Shutterstock

to select and invite bowl-eligible (a team that has six wins against its Division I-A opponents in that season) successful teams (as per the ratings and rankings) that will play in an exciting and competitive game, attract fans of both schools, and keep the remaining fans tuned in via a variety of media outlets for advertising.

In an unusual data mining study, Delen, Cogdell, and Kasap (2012) used eight years of bowl game data along with three popular data mining techniques (decision trees, neural networks, and support vector machines) to predict both the classification-type outcome of a game (win versus loss) as well as the regression-type outcome (projected point difference between the scores of the two opponents). What follows is a shorthand description of their study.

METHODOLOGY

In this research, Delen and his colleagues followed a popular data mining methodology called CRISP-DM (Cross-Industry Standard Process for Data Mining), which is a six-step process. This popular methodology, which is covered in detail in Chapter 5, provided them with a systematic and structured way to conduct the underlying data mining study and hence improved the likelihood of obtaining accurate and reliable results. To objectively assess the prediction power of the different model types, they used a cross-validation methodology, called k -fold cross-validation. Details on k -fold cross-validation can be found in Chapter 5. Figure 3.20 graphically illustrates the methodology employed by the researchers.

DATA ACQUISITION AND DATA PREPROCESSING

The sample data for this study is collected from a variety of sports databases available on the Web, including jhowel.net, ESPN.com, Covers.com, ncaa.org, and raunuzusstreet.com. The data set included 244 bowl games, representing a complete set of eight seasons of college football bowl games played between 2002 and 2009. We also included an out-of-sample data set (2010–2011 bowl games) for additional validation purposes. Exercising one of the popular data mining rules-of-thumb, they included as much relevant information into the model as possible. Therefore, after an in-depth variable identification and collection process, they ended up with a data set that included 36 variables, of which the first 6 were the identifying variables (i.e., name and the year of the bowl game, home and away team names and their athletic conferences—see variables 1–6 in Table 3.6), followed by 28 input variables (which included variables delineating a team's seasonal statistics on offense and defense, game outcomes, team composition characteristics, athletic conference characteristics, and how they fared against the odds—see variables 7–34 in 6), and finally the last two were the output variables (i.e., ScoreDiff—the score difference between the home team and the away team represented with an integer number, and WinLoss—whether the home team won or lost the bowl game represented with a nominal label).

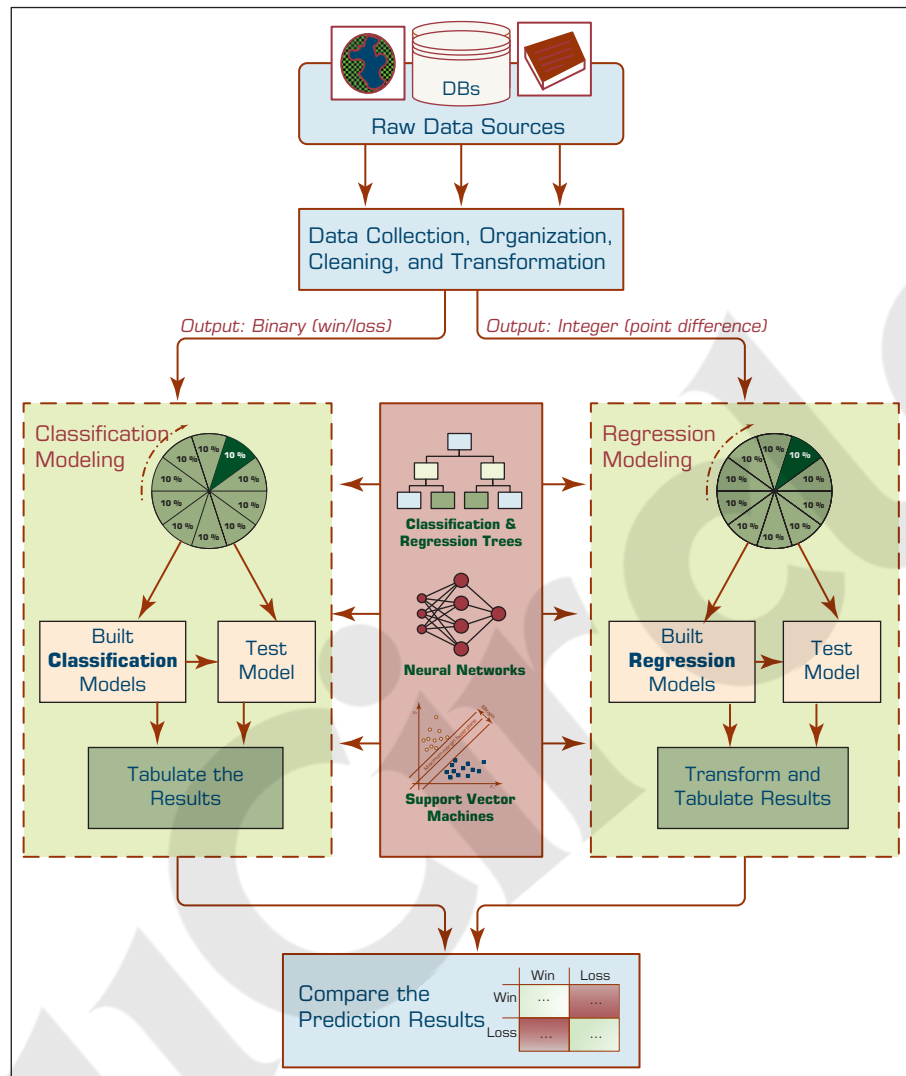


FIGURE 3.20 The Graphical Illustration of the Methodology Employed in the Study.

In the formulation of the data set, each row (a.k.a. tuple, case, sample, example, etc.) represented a bowl game, and each column stood for a variable (i.e., identifier/input or output type). To represent the game-related comparative characteristics of the two opponent teams, in the input variables, we calculated and used the differences between the measures of the home and away teams. All these variable values are calculated from the home team's perspective. For instance, the variable PPG (average number of points a team scored per game) represents the difference between the home team's PPG and away team's PPG. The output variables represent whether the home team wins or loses the bowl game. That is, if the ScoreDiff variable takes a positive integer number, then the home team is expected to win the game by that margin, otherwise (if the ScoreDiff variable takes a negative integer number) then the home team is expected to lose the game by that margin. In the case of WinLoss, the value of the output variable is a binary label, "Win" or "Loss" indicating the outcome of the game for the home team.

RESULTS AND EVALUATION

In this study, three popular prediction techniques are used to build models (and to compare them to each other): artificial neural networks, decision trees, and support vector machines. These prediction techniques are selected based on their capability of modeling both classification as

TABLE 3.6 Description of the Variables Used in the Study

No	Cat	Variable Name	Description
1	ID	YEAR	Year of the bowl game
2	ID	BOWLGAME	Name of the bowl game
3	ID	HOMETEAM	Home team (as listed by the bowl organizers)
4	ID	AWAYTEAM	Away team (as listed by the bowl organizers)
5	ID	HOMECONFERENCE	Conference of the home team
6	ID	AWAYCONFERENCE	Conference of the away team
7	I1	DEFPTPGM	Defensive points per game
8	I1	DEFYDPM	Defensive rush yards per game
9	I1	DEFYDPM	Defensive yards per game
10	I1	PPG	Average number of points a given team scored per game
11	I1	PYDPM	Average total pass yards per game
12	I1	RYDPM	Team's average total rush yards per game
13	I1	YRDPM	Average total offensive yards per game
14	I2	HMWIN%	Home winning percentage
15	I2	LAST7	How many games the team won out of their last 7 games
16	I2	MARGOVIC	Average margin of victory
17	I2	NCTW	Nonconference team winning percentage
18	I2	PREVAPP	Did the team appeared in a bowl game previous year
19	I2	RDWIN%	Road winning percentage
20	I2	SEASTW	Winning percentage for the year
21	I2	TOP25	Winning percentage against AP top 25 teams for the year
22	I3	TSOS	Strength of schedule for the year
23	I3	FR%	Percentage of games played by freshmen class players for the year
24	I3	SO%	Percentage of games played by sophomore class players for the year
25	I3	JR%	Percentage of games played by junior class players for the year
26	I3	SR%	Percentage of games played by senior class players for the year
27	I4	SEASOVUn%	Percentage of times a team went over the O/U* in the current season
28	I4	ATSCOV%	Against the spread cover percentage of the team in previous bowl games
39	I4	UNDER%	Percentage of times a team went under in previous bowl games
30	I4	OVER%	Percentage of times a team went over in previous bowl games

(Continued)

TABLE 3.6 (Continued)

No	Cat	Variable Name	Description
31	I4	SEASATS%	Percentage of covering against the spread for the current season
32	I5	CONCH	Did the team win their respective conference championship game
33	I5	CONFSOS	Conference strength of schedule
34	I5	CONFWIN%	Conference winning percentage
35	O1	ScoreDiff°	Score difference (HomeTeamScore – AwayTeamScore)
36	O2	WinLoss°	Whether the home team wins or loses the game

*Over/Under—Whether or not a team will go over or under of the expected score difference.

° Output variables—ScoreDiff for regression models and WinLoss for binary classification models.

I1: Offense/defense; I2: game outcome; I3: team configuration; I4: against the odds; I5: conference stats.

ID: Identifier variables; O1: output variable for regression models; O2: output variable for classification models.

well as regression-type prediction problems and their popularity in recently published data mining literature. More details about these popular data mining methods can be found in Chapter 5.

To compare predictive accuracy of all models to one another, the researchers used a stratified k -fold cross-validation methodology. In a stratified version of k -fold cross-validation, the folds are created in a way that they contain approximately the same proportion of predictor labels (i.e., classes) as the original data set. In this study, the value of k is set to 10 (i.e., the complete set of 244 samples are split into 10 subsets, each having about 25 samples), which is a common practice in predictive data mining applications. A graphical depiction of the 10-fold cross-validations was shown earlier in this chapter. To compare the prediction models that were developed using the aforementioned three data mining techniques, the researchers chose to use three common performance criteria: accuracy, sensitivity, and specificity. The simple formulas for these metrics were also explained earlier in this chapter.

The prediction results of the three modeling techniques are presented in Tables 3.7 and 3.8. Table 3.7 presents the 10-fold cross-validation results of the classification methodology where the three data mining techniques are formulated to have a binary-nominal output variable (i.e., *WinLoss*). Table 3.8 presents the 10-fold cross-validation results of the regression-based classification methodology, where the three data mining techniques are formulated to have a numerical output variable (i.e., *ScoreDiff*). In the regression-based classification prediction, the numerical output of the models is converted to a classification type by labeling the positive *WinLoss* numbers with a “Win” and negative *WinLoss* numbers with a “Loss,” and then tabulating them in the confusion matrixes. Using the confusion matrices, the overall prediction accuracy, sensitivity, and specificity of each model type are calculated and presented in these two tables. As the results indicate, the classification-type prediction methods performed better than regression-based classification-type prediction methodology. Among the three data mining technologies, classification and regression trees produced better prediction accuracy in both prediction methodologies. Overall, classification and regression tree classification models produced a 10-fold cross-validation accuracy of 86.48%, followed by support vector machines (with a 10-fold cross-validation accuracy of 79.51%) and neural networks (with a 10-fold cross-validation accuracy of 75.00%). Using a t -test, researchers found that these accuracy values were significantly different at 0.05 alpha level, that is, the decision tree is a significantly better predictor of this domain than the neural network and support vector machine, and the support vector machine is a significantly better predictor than neural networks.

The results of the study showed that the classification-type models predict the game outcomes better than regression-based classification models. Even though these results are specific to the application domain and the data used in this study, and therefore should not be generalized beyond the scope of the study, they are exciting because decision trees are not only the best predictors but also the best in understanding and deployment, compared to the other two machine-learning techniques employed in this study.

TABLE 3.7 Prediction Results for the Direct Classification Methodology

Prediction Method (Classification*)		Confusion Matrix		Accuracy** (in %)	Sensitivity (in %)	Specificity (in %)
		Win	Loss			
ANN (MLP)	Win	92	42	75.00	68.66	82.73
	Loss	19	91			
SVM (RBF)	Win	105	29	79.51	78.36	80.91
	Loss	21	89			
DT (C&RT)	Win	113	21	86.48	84.33	89.09
	Loss	12	98			

*The output variable is a binary categorical variable (Win or Loss); differences were sig (** $p < 0.01$).

TABLE 3.8 Prediction Results for the Regression-Based Classification Methodology

Prediction Method (Regression-Based*)		Confusion Matrix		Accuracy**	Sensitivity	Specificity
		Win	Loss			
ANN (MLP)	Win	94	40	72.54	70.15	75.45
	Loss	27	83			
SVM (RBF)	Win	100	34	74.59	74.63	74.55
	Loss	28	82			
DT (C&RT)	Win	106	28	77.87	76.36	79.10
	Loss	26	84			

*The output variable is a numerical/integer variable (point-diff); differences were sig (** $p < 0.01$).

Sources: Delen, D., Cogdell, D., & Kasap, N. (2012). A comparative analysis of data mining methods in predicting NCAA bowl outcomes. *International Journal of Forecasting*, 28, 543–552; Freeman, K. M., & Brewer, R. M. (2016). The politics of American college football. *Journal of Applied Business and Economics*, 18(2), 97–101.

Time Series Forecasting

Sometimes the variable that we are interested in (i.e., the response variable) may not have distinctly identifiable explanatory variables, or there may be too many of them in a highly complex relationship. In such cases, if the data is available in a desired format, a prediction model, the so-called time series, can be developed. A time series is a sequence of data points of the variable of interest, measured and represented at successive points in time spaced at uniform time intervals. Examples of time series include monthly rain volumes in a geographic area, the daily closing value of the stock market indexes, and daily sales totals for a grocery store. Often, time series are visualized using a line chart. Figure 3.21 shows an example time series of sales volumes for the years 2008 through 2012 in a quarterly basis.

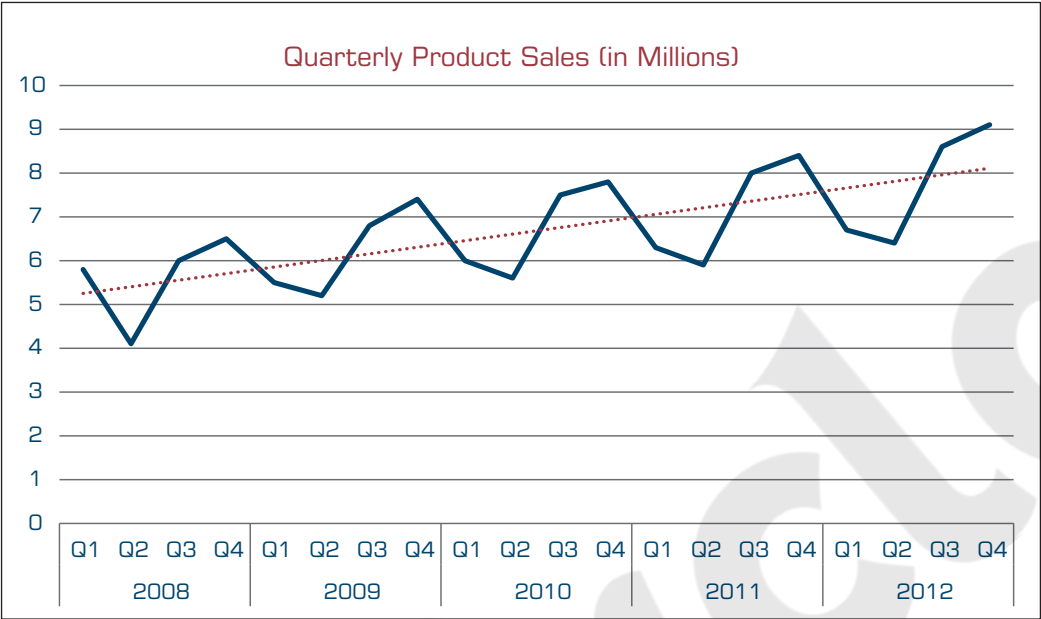


FIGURE 3.21 A Sample Time Series of Data on Quarterly Sales Volumes.

Time series forecasting is the use of mathematical modeling to predict future values of the variable of interest based on previously observed values. The time series plots/charts look and feel very similar to simple linear regression in that as was the case in simple linear regression, in time series there are two variables: the response variable and the time variable presented in a scatter plot. Beyond this look similarity, there is hardly any other commonality between the two. Although regression analysis is often employed in testing theories to see if current values of one or more explanatory variables explain (and hence predict) the response variable, the time series models are focused on extrapolating on their time-varying behavior to estimate the future values.

Time series forecasting assumes all the explanatory variables are aggregated and consumed in the response variable’s time-variant behavior. Therefore, capturing of the time-variant behavior is the way to predict the future values of the response variable. To do that the pattern is analyzed and decomposed into its main components: random variations, time trends, and seasonal cycles. The time series example shown in Figure 3.21 illustrates all these distinct patterns.

The techniques used to develop time series forecasts range from very simple (the naïve forecast that suggests today’s forecast is the same as yesterday’s actual) to very complex like ARIMA (a method that combines autoregressive and moving average patterns in data). Most popular techniques are perhaps the averaging methods that include simple average, moving average, weighted moving average, and exponential smoothing. Many of these techniques also have advanced versions where seasonality and trend can also be taken into account for better and more accurate forecasting. The accuracy of a method is usually assessed by computing its error (calculated deviation between actuals and forecasts for the past observations) via mean absolute error (MAE), mean squared error (MSE), or mean absolute percent error (MAPE). Even though they all use the same core error measure, these three assessment methods emphasize different aspects of the error, some penalizing larger errors more so than the others.